

STRUCTURAL GROUP CHAIRMAN FACTUAL REPORT
(82 pages)



**NATIONAL TRANSPORTATION SAFETY BOARD
OFFICE OF HIGHWAY SAFETY
WASHINGTON, DC 20594**

STRUCTURAL GROUP CHAIRMAN FACTUAL REPORT

A. ACCIDENT

LOCATION: Eastbound Interstate 90 connector tunnel, about 1000 feet west of the Ted Williams Tunnel
Boston, Suffolk County, Massachusetts

VEHICLE: 1991 Buick 4-door passenger car

MOTOR CARRIER: N/A (Privately Owned)

DATE: Monday July 10, 2006

TIME: 11:01 p.m. Local Time

FATALITIES: 1

INJURIES: 1

CASE NR: HWY-06-MH-024

B. ACCIDENT SUMMARY

About 11:01 p.m. local time on Monday, July 10, 2006, a 1991 Buick passenger car was traveling eastbound in the left lane of the Interstate 90 connector tunnel to the Ted Williams Tunnel in Boston, Massachusetts. The vehicle was occupied by a 46-year-old male driver and a 38-year-old female passenger. A section of the tunnel's ceiling structure, consisting of ten 8-foot by 12-foot concrete panels and associated hardware, collapsed and struck the Buick. The Ceiling was held in place by multiple stainless steel anchors epoxied to the roof of the tunnel. A post-crash inspection revealed that 20 of the anchors had pulled out from the roof of the tunnel. These anchors were used to attach eight metal brackets that supported a portion of the ceiling structure. Initially, two of the concrete panels fell onto the passenger vehicle crushing the right side of the roof. Following this impact, the passenger car came to rest against the north wall of the tunnel, and additional panels came to rest on the vehicle. As a result, the female passenger occupying the front right seat was fatally injured and the male driver received minor injuries.

C. DETAILS OF THE INVESTIGATION

A description and map was obtained of the entire Central Artery / Tunnel (CA/T) Project at the time of completion. Two illustrations were developed showing a three dimensional view of the ceiling system used in the I-90 connector tunnel and a cross section showing the eastbound travel lanes of the I-90 connector tunnel.

An inventory of the concrete ceiling module was done that identified the properties of the module as shown on the approved record drawings and from a survey conducted by NTSB investigators on July 17, 2006. An inventory was done of the spacing of the rod hanger pin assembly along the panel supports. The properties of the concrete ceiling module were weighed and documented on November 28, 2006. An inventory was taken of the adhesive anchors and holes.

Highway data was obtained that included average daily traffic, vehicle classification data, traffic and fatal accident data, functional classification, highway design, and speed limit.

Research was done to document the construction history of the D Street Portal (West Tunnel). Documentation was made of different construction contracts of the I-90 connector tunnel and Ted Williams Tunnel. Research was done to document key recommendations from a Value Engineering Report – Tunnel Finishes dated March 1995. The contractor provided a progress schedule of the construction of the I-90 connector tunnel finishes contract.

The section design consultant provided to NTSB investigators a brief summation of the design principles and relevant loading of the adhesive anchors. Documentation was made of the contractor submitting an anchor adequacy submittal stamped and sealed by a Professional Engineer (PE). Documentation was made of a Design Policy Memorandum No. 107 dated March 13, 1997. The Federal Highway Administration (FHWA) Turner-Fairbank Highway Research Center developed a finite element analysis to determine the loading conditions and structural redundancy of the ceiling system. Research was done to determine the project

participant responsibilities. Research was done to determine the design of suspended ceiling panels in other tunnels throughout the country.

Documentation was made of the installation procedure for the concrete ceiling panels. Research was done that included providing a detailed description of the tunnel ventilation system in the Central Artery / Tunnel (CA/T) Project.

Research was done to document Power Fasteners, Inc. guidelines for installation of adhesive anchors and ultimate and allowable load capacities. Research was done to determine the American Association of State Highway and Transportation Officials (AASHTO) guidelines on the installation, inspection, and testing of embedment anchors.

Documentation was made of 15 correspondence documents between Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, and Modern Continental Company (MCC), the tunnel finishes contractor, regarding a small percentage of adhesive anchors that showed signs of tensile movement in the High Occupancy Vehicle (HOV) ceiling mockup area.

Research was done to document a December 1998 report from the Commonwealth of Massachusetts Office of the Inspector General regarding the use of anchor bolts in the Ted Williams Tunnel.

Documentation was made of a proposal made by MCC for a lighter ceiling system. Documentation was made of a contractor submittal for the use of screen tubes.

Research was done to determine if any inspections were performed to the ceiling system from the time the I-90 eastbound connector tunnel was opened to traffic to the day of the collapse. Documentation was made of a scope of services to develop a Structural Inspection Program. Documentation was made of an inspection manual for tunnels and boat structures prepared by B/PB. Documentation was made of the status of tunnel inspections the Massachusetts Turnpike Authority (MTA) had instituted after the day of the collapse. Research was done to determine the inspection standards recommended in Federal Highway Administration's (FHWA) "Highway and Rail Transit Tunnel Inspection Manual" and the national bridge inspection program.

Research was done to determine if any national standards exist for the design of tunnel finishes in FHWA's "Road Tunnel Design Guidelines".

Documentation was made regarding an alternate anchor system to support the ceiling system proposed by the section design consultant. Documentation was made of a memorandum citing direction to use a factor of safety of two applied to manufacturer's ultimate capacities to arrive at allowable capacities for adhesive design.

Documentation was made of emergency legislation issued by Governor Mitt Romney that established a Metropolitan Highway System (MHS) Safety Review Team. The Safety Review Team prepared an eleven-volume series of reports entitled Stem to Stern Safety Review – Phase 1 of the MHS tunnels and facilities.

1. PREFATORY DATA

1.1 ACCIDENT LOCATION

The accident occurred in the eastbound travel lanes of the Interstate 90 (I-90) connector tunnel at Mile Marker 135.25. The I-90 connector tunnel is located immediately west of the Ted Williams tunnel that connects to the Logan International Airport. Figure 1 illustrates the accident site was located in the City of Boston in Suffolk County, Massachusetts.

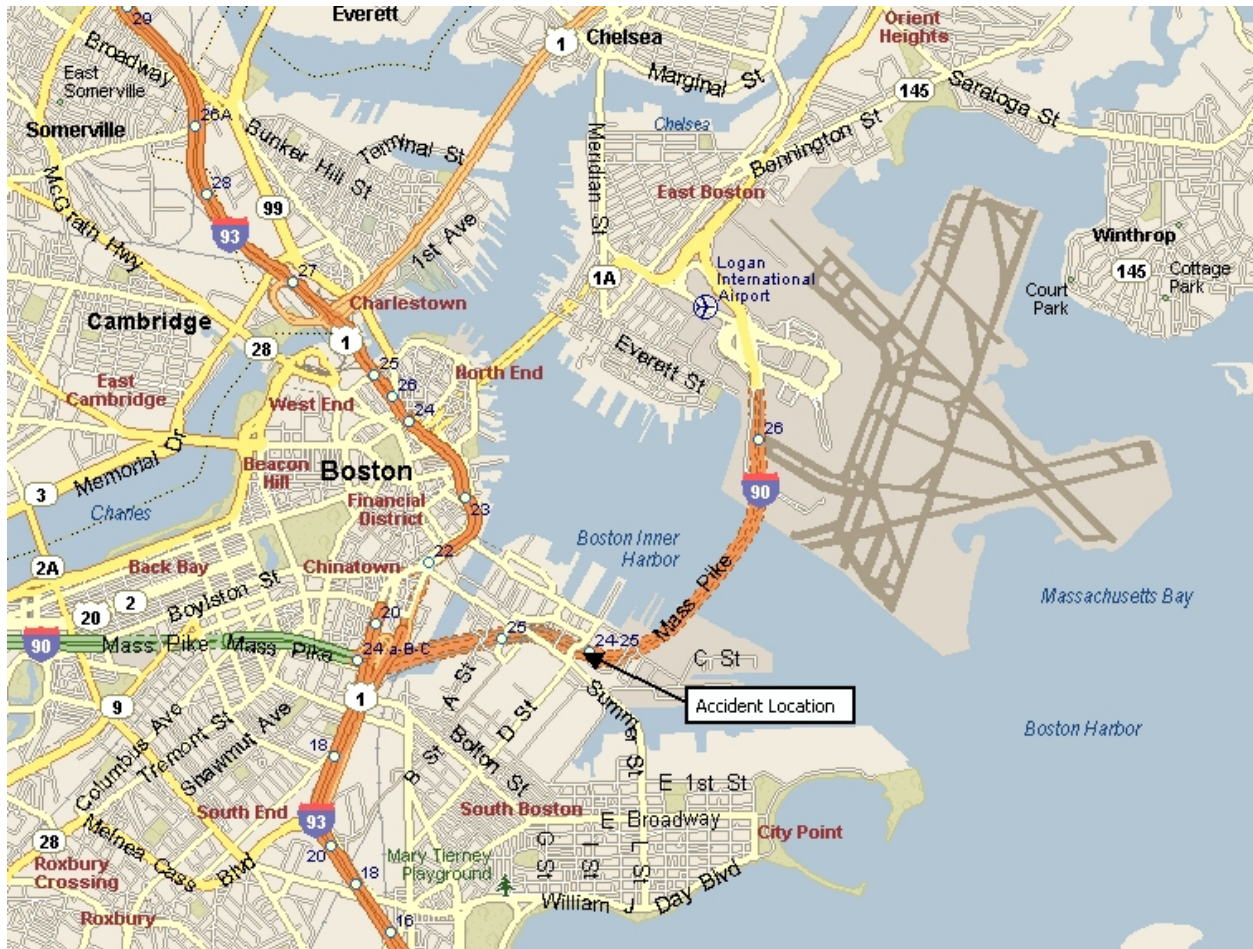


Figure 1 – Location Map

1.2 WEATHER DATA

The closest official National Weather Service (NWS) weather reporting facility to the accident site was from Boston Logan International Airport (KBOS) located approximately 1 mile east of the accident site. The airport reported the following weather conditions on July 10, 2006 at 10:54 p.m. local time: wind from the southwest at 10 miles per hour (mph), visibility unrestricted at 10 statute miles, and a few clouds at 7,000 feet (mostly sunny).

Table 1 shows the low temperature, high temperature, and precipitation reported on July 10, 2006 and six days prior to the accident:

Table 1 – Weather conditions

Date	Low Temperature	High Temperature	Precipitation Reported
July 10, 2006	66 degrees F	79 degrees F	No precipitation reported
July 9, 2006	63 degrees F	88 degrees F	No precipitation reported
July 8, 2006	70 degrees F	75 degrees F	No precipitation reported
July 7, 2006	63 degrees F	81 degrees F	No precipitation reported
July 6, 2006	64 degrees F	77 degrees F	0.15 inches of precipitation reported
July 5, 2006	73 degrees F	81 degrees F	No precipitation reported
July 4, 2006	68 degrees F	88 degrees F	Less than 0.01 inches of precipitation reported

2. OVERVIEW OF DOWNTOWN TUNNEL SYSTEM

2.1 DESCRIPTION AND MAP OF ENTIRE TUNNEL SYSTEM

Figure 2 illustrates the Central Artery / Tunnel (CA/T) Project at the time of completion in 2004. The CA/T Project was designed to accommodate approximately 2.5 times more vehicles per day than the original elevated artery. The CA/T Project was designed to accommodate approximately 190,000 vehicles per day while the original elevated artery accommodated approximately 75,000 vehicles per day. The CA/T Project consisted of 161 lane miles of highway in a 7.5-mile corridor. The CA/T Project included 5 miles of tunnel, 6 interchanges, and 200 bridge structures. The north-south artery, I-93, was an 8 to 10 lane, limited access, underground expressway that replaced the original 6-lane elevated highway. The east-west artery, I-90, was a new tunnel under South Boston that extended under Boston Inner Harbor to Logan International Airport. The CA/T Project claimed the world's widest asymmetrical cable-stayed bridge (Zakim Bridge) and the world's largest highway tunnel ventilation system.



Figure 2 – Central Artery / Tunnel Project Map at Completion
(Courtesy of the Massachusetts Turnpike Authority)

2.2 ILLUSTRATION SHOWING THREE DIMENSIONAL VIEW OF CEILING SYSTEM

Figure 3 illustrates a three dimensional view of the ceiling system used in the I-90 connector tunnel. The ceiling system consisted of concrete ceiling panels that measured 12 feet wide by 8 feet long by 4 inches thick. A ceiling module was referred to as five (5) concrete ceiling panels that were located side by side and oriented parallel to the travel lanes. The total length of a ceiling module was 40 feet.

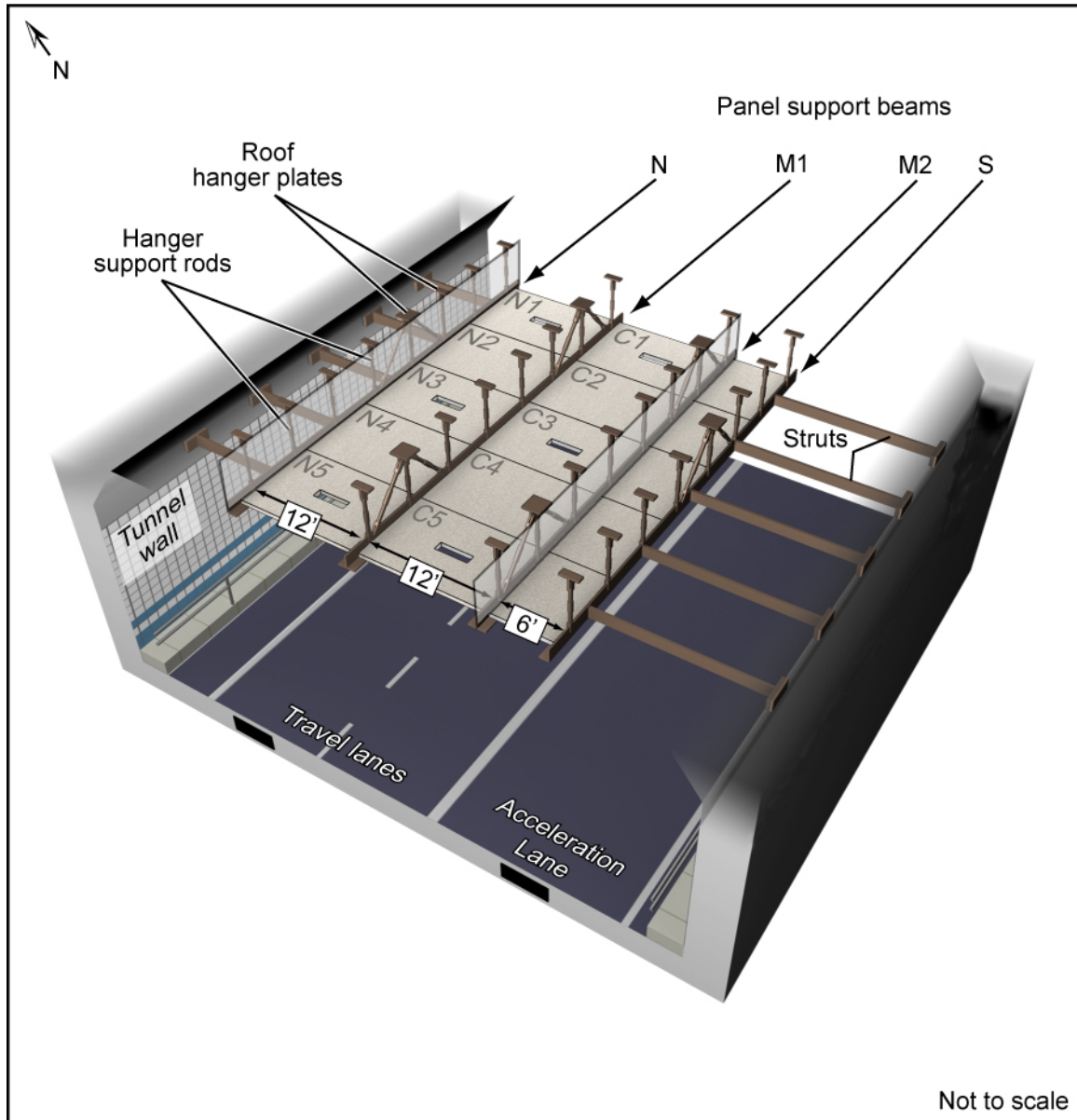


Figure 3 – Three dimensional view of the ceiling system used in the I-90 connector tunnel

The structural members used to support the concrete ceiling panels were panel supports, hanger supports, and adhesive anchors embedded into the tunnel ceiling. The vertical hanger supports and adhesive anchors worked in pure tension. The angled (45 degree) hanger supports prevented movement of the ceiling system in the longitudinal direction (parallel to the travel lanes) while the struts prevented movement of the ceiling system in the transverse direction.

The portion of the ceiling system that collapsed on July 10, 2006 was the right middle row that included the ten concrete ceiling panels adjacent to the panel support. Each concrete ceiling panel weighed approximately 4,865 pounds or 2.4 tons. The total weight of the concrete ceiling panels and structural members that collapsed was approximately 52,102 pounds or 26 tons.

2.3 ILLUSTRATION SHOWING CROSS SECTION OF EASTBOUND TRAVEL LANES OF I-90 CONNECTOR TUNNEL

Figure 4 illustrates a cross section of the eastbound travel lanes of the I-90 connector tunnel. The exhaust duct shaft is located above the roadway pavement between the concrete ceiling panels and the tunnel ceiling. The supply air duct shafts are located below the roadway pavement.

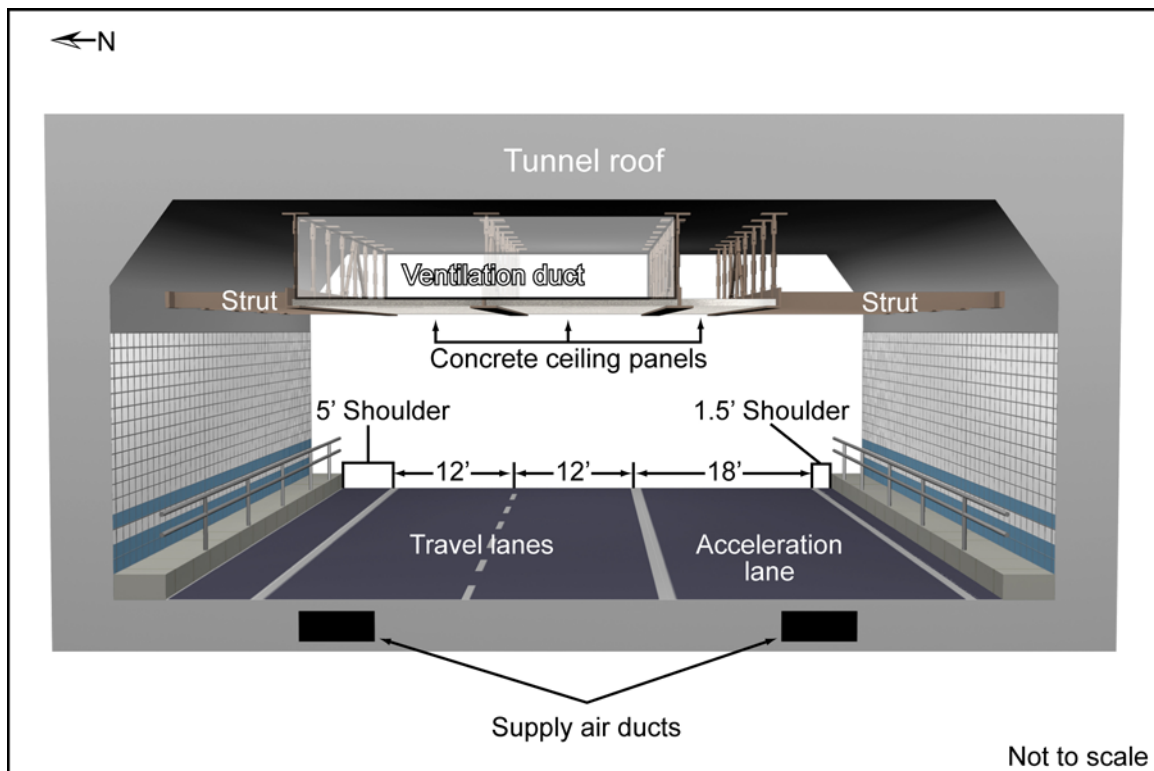


Figure 4 – Cross section of eastbound travel lanes of the I-90 connector tunnel

The 12-foot wide concrete ceiling panels are located directly overhead and in line with the travel lanes for aesthetic considerations. The vertical distance from the roadway pavement to the concrete ceiling panels was approximately 17 feet. The vertical distance from the concrete ceiling panels to the tunnel ceiling was approximately 5 ½ feet. The width of the entire cross section of roadway was approximately 48.5 feet. The cross section of roadway consisted of a 5-foot wide left shoulder, two 12-foot wide travel lanes, an 18-foot wide acceleration lane, and a 1.5-foot wide right shoulder.

3. CEILING MODULE FRAMING

3.1 CEILING MODULE CHARACTERISTICS

The type of concrete module that fell in the eastbound lanes of I-90 was referred to as module group type 5¹. The properties of the concrete ceiling module as shown on the approved record drawings are shown in Table 2.

Table 2 – Properties of the concrete ceiling module as shown on the approved record drawings (module group type 5)

NUMBER OF PANELS	
10 panels (Collapsed)	12' x 8' x 4"
5 panels	6' x 8' x 4"
PANEL SUPPORTS	
North and South End (39' length)	WT 13.5 x 42 (Angled stiff plate on strut side)
Hold down angle (39' length)	L 5 x 3 1/2
Bolts through concrete panel	5/8" diameter welded stud bolt and pipe sleeve
Middle – North and South End (39' length) (Middle – North End Collapsed)	Plate ¾ x 12, L 8 x 6 x ½ (2)
Hold down angles (39' length)	L 5 x 3 ½ (2)
Bolts through concrete panel	5/8" diameter welded stud bolt and pipe sleeve
ROD HANGER ASSEMBLY	
Vertical rod hanger (32 total, 8 per row, 4'-6" length) (8 collapsed in middle – north end)	1" diameter threaded rod and 6" turnbuckle
Pin connection	No. 3 clevis with holes for 1 ½" diameter pin
Angled (45 degrees) rod hanger (12 total, 5'-8" length)	1" diameter threaded rod and

¹Massachusetts Highway Department, Central Artery (I-93)/Tunnel (I-90) Project, Plan and Profile of Highways, City of Boston, Suffolk County, Federal Aid Project STP 90-1 (216), Contract No. C09B2, I-90 Seaport Access Tunnel Finishes, Record Drawings Approved 6/20/05, Drawing No. S-013.

(4 collapsed in middle – north end)	6” turnbuckle
Pin connection	No. 3 clevis with holes for 1 ½” diameter pin
HANGER SUPPORT	
2 bolt connection (26 total) (6 collapsed in middle – north end)	WT 6
4 bolt connection (6 total) (2 collapsed in middle – north end)	WT 6 x 48
Adhesive anchors (76 total) (20 collapsed in middle – north end)	5/8” diameter adhesive anchors 5” minimum embedment depth
CEILING STRUT	
Ceiling strut	WT 5

Table 3 shows the properties of the concrete ceiling module as found in a survey conducted on July 17, 2006 at the Massachusetts State Police - Troop E Barracks located at 50 Massport Haul Road in Boston, Massachusetts.

Table 3 – Properties of the concrete ceiling module as found in a survey conducted on July 17, 2006 at the Massachusetts State Police – Troop E Barracks

NUMBER OF PANELS	
10 panels ² (Collapsed)	12’ x 8’ x 4”
5 panels	6’ x 8’ x 4”
PANEL SUPPORTS	
North and South End (39’ length)	WT 13.5 x 42 (Rectangular stiff plate on strut side)
Hold down U channel	4” x 4” bolted to web
Bolts through concrete panel	5/8” diameter welded stud bolt and pipe sleeve
Middle – North and South End (39’ length) (Middle – North End Collapsed)	Plate ¾ x 12, L 8 x 6 x ½ (2)
Hold down plate	4” x 4” x 3/8”
Bolts through concrete panel	5/8” diameter welded stud bolt and pipe sleeve
ROD HANGER ASSEMBLY	
Vertical rod hanger (32 total, 8 per row, 4’-6” length) (8 collapsed in middle – north end)	1” diameter threaded rod and 6” turnbuckle
Pin connection	No. 3 clevis with holes for 1 ½” diameter pin

²Concrete ceiling panels have a 3.4’ x 0.5’ exhaust port opening located in the middle of the panel. The panels with exhaust port openings are spaced approximately every other panel.

Angled (45 degrees) rod hanger (12 total, 5'-8" length) (4 collapsed in middle – north end)	1" diameter threaded rod and 6" turnbuckle
Pin connection	No. 3 clevis with holes for 1 ½" diameter pin
HANGER SUPPORT	
2 bolt connection (26 total) (6 collapsed in middle – north end)	WT 6
4 bolt connection (6 total) (2 collapsed in middle – north end)	WT 6 x 48
Adhesive anchors (76 total) (20 collapsed in middle – north end)	5/8" diameter adhesive anchors Embedded at various depths
CEILING STRUT	
Ceiling strut	WT 5

3.2 SPACING OF ROD HANGER PIN ASSEMBLY ALONG PANEL SUPPORTS

Table 4 shows the spacing of the rod hanger pin assembly measured along the M1 (middle – north end) panel support that collapsed. The spacing of the rod hanger pin assembly was measured from the west end of the panel support.

Table 4 – Spacing of the rod hanger pin assembly measured along the M1 (middle – north end) panel support that collapsed (measurements taken from west end of panel support)

Rod hanger pin assembly	Spacing from west end of panel support
Vertical rod hanger	3'-0"
Angled (45 degrees) rod hanger	3'-6"
Vertical rod hanger	8'-0"
Angled (45 degrees) rod hanger	12'-6"
Vertical rod hanger	13'-0"
Vertical rod hanger	17'-6"
Vertical rod hanger	23'-0"
Vertical rod hanger	27'-6"
Angled (45 degrees) rod hanger	28'-0"
Vertical rod hanger	32'-6"
Angled (45 degrees) rod hanger	37'-0"
Vertical rod hanger	37'-6"
East end of panel support	39'-0"

The rod hanger assembly for the M1 (middle – north end) panel support consisted of 2-four bolt hanger support connections and 6-two bolt hanger support connections. The number and spacing of the rod hanger pin assembly for the M2 (middle – south end) panel support was identical to the M1 (middle – north end) panel support.

The rod hanger assembly for the north and south end panel supports consisted of 1-four bolt hanger support connection and 7-two bolt hanger support connections.

3.3 WEIGHT OF CONCRETE CEILING MODULE PROPERTIES

The properties of the concrete ceiling module were weighed³ at the D Street McCourt Construction Yard on Tuesday, November 28, 2006. Table 5 shows the weight of the concrete ceiling module properties.

Table 5 – Weight of the concrete ceiling module properties

Concrete ceiling module properties	Weight
12' x 8' x 4" concrete ceiling panel without exhaust port opening	4,865 pounds
12' x 8' x 4" concrete ceiling panel with exhaust port opening	4,640 pounds
6' x 8' x 4" concrete ceiling panel without exhaust port opening	2,483 pounds
Plate $\frac{3}{4}$ x 12, L 8 x 6 x $\frac{1}{2}$ (2) panel support	3,200 pounds
WT 13.5 x 42 panel support	1,800 pounds
2 bolt hanger support connection	46 pounds
4 bolt hanger support connection	163 pounds
WT 5 ceiling strut (eastbound lanes north side, 9' length)	150 pounds
WT 5 ceiling strut (eastbound lanes south side, 20' length)	450 pounds
1 corrugated metal exhaust duct closure panel (5'-6" height and 2'-0" length)	50 pounds

³The weight of the concrete ceiling panels, both with and without the exhaust port opening, was measured using a load cell provided by the Federal Highway Administration. The weight of the built-up steel panel supports was based on calculations using a standard cross section for beam elements and measurements of the length of the element. NTSB investigators measured the weight of the hanger assemblies. The 2-anchor hanger support connection consists of the ceiling hanger support, the rod hanger with turnbuckle, and the pins used to secure the hanger at each end. The 4-anchor hanger support connection consists of the ceiling hanger support, one vertical and two 45-degree rod hangers with turnbuckles, and the pins used to secure the hangers at each end.

Table 6 shows the approximate weight of the concrete ceiling module properties that collapsed.

Table 6 – Weight of the concrete ceiling module properties that collapsed

Concrete ceiling module properties	Quantity	Weight	Total Weight
12' x 8' x 4" concrete ceiling panel without exhaust port opening	4	4,865 pounds	19,460 pounds
12' x 8' x 4" concrete ceiling panel with exhaust port opening	6	4,640 pounds	27,840 pounds
Plate $\frac{3}{4}$ x 12, L 8 x 6 x $\frac{1}{2}$ (2) panel support	1	3,200 pounds	3,200 pounds
2 bolt hanger support connection	6	46 pounds	276 pounds
4 bolt hanger support connection	2	163 pounds	326 pounds
1 corrugated metal exhaust duct closure panel	20	50 pounds	1,000 pounds
Approximate total weight of concrete ceiling module properties that collapsed			52,102 pounds or 26 tons

Table 7 shows the approximate weight of the entire concrete ceiling group type 5 module.

Table 7 – Weight of the entire concrete ceiling group type 5 module

Concrete ceiling module properties	Quantity	Weight	Total Weight
12' x 8' x 4" concrete ceiling panel without exhaust port opening	4	4,865 pounds	19,460 pounds
12' x 8' x 4" concrete ceiling panel with exhaust port opening	6	4,640 pounds	27,840 pounds
6' x 8' x 4" concrete ceiling panel without exhaust port opening	5	2,483 pounds	12,415 pounds
Plate $\frac{3}{4}$ x 12, L 8 x 6 x $\frac{1}{2}$ (2) panel support	2	3,200 pounds	6,400 pounds
WT 13.5 x 42 panel support	2	1,800 pounds	3,600 pounds
2 bolt hanger support connection	26	46 pounds	1,196 pounds
4 bolt hanger support connection	6	163 pounds	978 pounds
WT 5 ceiling strut (eastbound lanes north side, 9' length)	5	150 pounds	750 pounds
WT 5 ceiling strut (eastbound lanes south side, 20' length)	5	450 pounds	2,250 pounds
1 corrugated metal exhaust duct closure panel	40	50 pounds	2,000 pounds
Approximate total weight of the entire concrete ceiling group type 5 module			76,889 pounds or 38 tons

3.4 INVENTORY OF ADHESIVE ANCHORS AND HOLES

Table 8 shows an inventory of the adhesive anchors and holes used to attach the hanger support connection to the ceiling of the tunnel.

Table 8 – Inventory of adhesive anchors and holes used to attach the hanger support connection to the ceiling of the tunnel

Diameter of Anchor (inches)	Length of Anchor (inches)	Type of Anchor	Diameter of Holes (inches)	Depth of Hole (inches)
5/8	8	3/16 grade stainless steel threaded anchor	3/4	5 to 5 ¼

4. HIGHWAY DATA

4.1 AVERAGE DAILY TRAFFIC

The average daily traffic (ADT) in the eastbound and westbound direction of I-90 in the vicinity of the accident was provided by the Massachusetts State Police. The average daily traffic is shown in Table 9.

Table 9 – Average daily traffic (ADT) in the eastbound and westbound direction of I-90 in the vicinity of the accident

Year	Average Daily Traffic (ADT) – Eastbound Direction	Average Daily Traffic (ADT) – Westbound Direction
1996	6,062 vehicles per day	7,396 vehicles per day
1997	9,065 vehicles per day	10,712 vehicles per day
1998	10,732 vehicles per day	14,173 vehicles per day
1999	12,171 vehicles per day	17,677 vehicles per day
2000	12,693 vehicles per day	17,763 vehicles per day
2001	11,248 vehicles per day	16,033 vehicles per day
2002	11,007 vehicles per day	16,880 vehicles per day
2003	31,881 vehicles per day	26,281 vehicles per day
2004	41,500 vehicles per day	40,300 vehicles per day
2005	43,000 vehicles per day	44,500 vehicles per day

4.2 VEHICLE CLASSIFICATION DATA

The Massachusetts Turnpike Authority (MTA) provided the vehicle classification data on I-90 eastbound in the vicinity of the accident. Vehicle classification data for I-90 eastbound had been derived from 2005 traffic counts and I-90 westbound vehicle classification counts at Toll Plazas 31 and 37 in East Boston. Table 10 summarizes the vehicle classification data that was collected by MTA.

Table 10 – Vehicle classification data on I-90 eastbound in the vicinity of the accident

Vehicle Classification	Percent (%)⁴	Average Weekday Traffic (AWDT)⁵
Two axle vehicles	97.6%	40,491
Three axle vehicles	1.2%	500
Four axle vehicles	0.3%	130
Five axle vehicles	0.9%	352
Six axle vehicles	0%	24
Seven axle vehicles	0%	3
Totals	100%	41,500

4.3 TRAFFIC AND FATAL ACCIDENT SUMMARY

The Massachusetts State Police provided the traffic and fatal accident summary in the eastbound and westbound direction of I-90 from Mile Marker 133 to 137 from 2003 through 2006. Table 11 provides the traffic and fatal accident summary.

Table 11 – Traffic and fatal accident summary in the eastbound and westbound direction of I-90 from Mile Marker 133 to 137 from 2003 through 2006

Mile Marker	I-90 Eastbound Direction			I-90 Westbound Direction			Total
	Property Damage	Personal Injury	Fatal	Property Damage	Personal Injury	Fatal	
2003							
133	9	7	0	10	1	0	27
133.5	1	1	0	2	0	0	4
134	11	3	0	4	2	0	20
134.5	0	1	0	2	0	0	3
135.0	6	1	0	3	0	0	10
135.5	0	0	0	0	0	0	0
136.0	0	0	0	0	0	0	0
136.5	1	0	0	0	0	0	1
137.0	0	0	0	0	0	0	0

⁴Based on June 2006 vehicle classification counts at Toll Plazas 31 and 37 (I-90 westbound) in East Boston.

⁵Based on 2005 I-90 eastbound traffic counts conducted by MTA Central Transportation Planning Staff (CTPS).

	28	13	0	21	3	0	65
2004							
133	10	0	0	11	4	0	25
133.5	2	0	0	1	0	0	3
134	12	2	0	8	1	0	23
134.5	3	3	0	5	1	0	12
135.0	5	3	0	4	1	0	13
135.5	2	0	0	3	0	0	5
136.0	2	1	0	3	0	0	6
136.5	0	0	0	1	2	0	3
137.0	1	1	0	1	0	0	3
	37	10	0	37	9	0	93
2005							
133	6	2	0	5	5	0	18
133.5	1	0	0	2	0	0	3
134	8	2	0	2	2	0	14
134.5	1	0	0	1	1	0	3
135.0	2	1	0	3	4	0	10
135.5	0	1	0	0	0	0	1
136.0	3	0	0	4	0	1	8
136.5	0	0	0	4	0	0	4
137.0	4	3	0	6	3	0	16
	25	9	0	27	15	1	77
2006							
133	3	1	0	3	0	0	7
133.5	1	0	0	0	0	0	1
134	3	0	0	3	1	0	7
134.5	0	0	0	1	0	0	1
135.0	2	0	0	3	1	0	6
135.5	0	1	0	0	0	0	1
136.0	0	0	0	0	0	0	0
136.5	0	0	0	0	1	0	1
137.0	3	0	0	1	0	0	4
	12	2	0	11	3	0	28

4.4 FUNCTIONAL CLASSIFICATION

The Massachusetts Turnpike Authority (MTA) functionally classified the Interstate 90 (I-90) connector tunnel as an urban principal arterial road. The American Association of State Highway and Transportation Officials (AASHTO) classified an urban principal arterial road⁶ as follows:

⁶A Policy on Geometric Design of Highways and Streets, American Association of State Highway and Transportation Officials (AASHTO), 2004, Fifth Edition, page 11.

“The urban principal arterial system serves the major centers of activity of urbanized areas, the highest traffic volume corridors, and the longest trip desires and carries a high proportion of the total urban area travel even though it constitutes a relatively small percentage of the total roadway network. The system should be integrated both internally and between major rural connections.”

4.5 HIGHWAY DESIGN

The eastbound lanes of the Interstate 90 (I-90) connector tunnel in the vicinity of the accident consisted of two travel lanes and an acceleration lane. The width of each travel lane was 12 feet and the width of the acceleration lane was 18 feet. A 5 foot wide shoulder was located left of the travel lanes. A 6 inch wide solid yellow line separated the left travel lane from the shoulder. A 1.5 foot wide shoulder was located right of the acceleration lane. The total paved area of the eastbound lanes of the I-90 connector tunnel in the vicinity of the accident was approximately 48.5 feet.

A raised walkway was located on both sides of the connector tunnel. The raised walkway located left of the shoulder was approximately 2.5 feet higher than the elevation of the roadway slab and was approximately 3 feet wide. A handrail was located at the edge of the raised walkway that was approximately 3.5 foot high. The raised walkway located right of the acceleration lane was approximately 3 feet higher than the elevation of the roadway slab and was approximately 2.5 feet wide.

4.6 SPEED LIMIT

The speed limit for the Interstate 90 (I-90) connector tunnel in the vicinity of the accident was 45 miles per hour (mph).

5. **CONSTRUCTION HISTORY**

The sequence of construction for the entire Central Artery / Tunnel (CA/T) Project was developed based on the early opening of the Ted Williams Tunnel to provide an “early” second link from Boston under Boston Harbor to the Logan International Airport. By completing this leg early a significant volume of traffic heading to Logan International Airport from the west and south of Boston could be redirected to this route effectively taking cars off the elevated expressway that would otherwise travel through downtown Boston to get to Logan International Airport via the Callahan Tunnel.

Completion of the Ted Williams Tunnel and the connections to South Boston and East Boston was called the “Early Opening”. There were three main contracts that composed the “Early Opening” that included C04A2 (Boston Marine Industrial Park Tunnel), C05A1 (Immersed Tube Tunnel), and C07A1 (Bird Island Flats Tunnel) located near Logan International Airport. These tunnels were completed to support the early opening of I-90 (partial) on December 15, 1995.

At both ends of the early opening contracts, temporary facilities were constructed to allow traffic to enter and exit the tunnels while the permanent facilities connecting to either end of the early opening contracts were completed.

In South Boston, the temporary facility consisted of a temporary ramp that connected D Street at surface street level to the Ted Williams Tunnel. The temporary ramp was needed for traffic to enter and exit the Ted Williams Tunnel to support the early opening. On the west end of the C04A2 contract, an approximate 200-foot section of permanent tunnel was designed and constructed. This section of tunnel was not intended to carry traffic at the time of the early opening. The 200-foot section of tunnel was identified as the “West Tunnel” on the C04A2 contract.

The West Tunnel served two purposes in the construction staging for the ultimate completion of the I-90 extension. First, the northern half of the West Tunnel provided a structure in which a temporary ramp could be constructed to connect D Street to the Ted Williams Tunnel to support the early opening. Second, the length of the tunnel was long enough so that temporary traffic detours could remain in place while construction to extend the tunnels to the west could occur and ultimately connect to I-93.

The West Tunnel was built to the same criteria as the rest of the tunnels in the C04A2 contract. It was designed and constructed in accordance with the design parameters in place at the time. It was anticipated the same tunnel finishes and ceiling system installed in the early opening tunnel sections would be designed and installed in the West Tunnel at some later date.

Table 12 shows the different construction contracts of the I-90 connector tunnel and the Ted Williams Tunnel. The I-90 connector tunnel was divided into three construction contracts. To reduce interfaces and ensure consistency, the tunnel finish design of the I-90 connector tunnel was consolidated into one construction contract.

Table 12 – Different construction contracts of the I-90 connector tunnel and the Ted Williams Tunnel

Construction Contract Number and Name	Section Design Consultant (SDC)	Contractor	Description
I-90 Connector Tunnel			
C09B1 – I-90 Fort Point Channel Crossing (West of the accident location)	Gannett-Fleming (6/93 - 2/98)	Modern Continental Construction (3/97 – 5/02)	The C09B1 construction contract includes the construction of cut-and-cover tunnels of approximately 745 linear feet of Ramp L tunnel, 640 linear feet of HOV eastbound tunnel and boat, 180 linear feet of I-90 eastbound tunnel and 145 linear feet of I-90 westbound tunnel west of the Fort Point Channel. It also includes the construction and installation of six concrete immersed tube

			tunnel sections totaling approximately 1,100 linear feet under the Fort Point Channel. Also in the scope is 425 linear feet of cut-and-cover tunnel east of the Fort Point Channel.
C01A3 – I-90 South Boston Interchange (East) (Immediately west of the accident location)	HDR Engineering, Inc. (5/93 – 3/97)	Kiewit / Atkinson / Cashman (1/97 – 7/02)	The C01A3 construction contract includes construction of a portion of I-90 in the South Boston area from the South Boston Haul Road to the C04A2 limit 200 feet west of D Street. The work consists of approximately 1,600 linear feet of varying width cut-and-cover reinforced concrete highway tunnel and boat sections, 1,600 linear feet of steel and precast box beam bridges, and 9,700 linear feet of surface roadway. In addition, scope includes the MBTA World Trade Center Station including approximately 1,200 linear feet of tunnel and boat section.
C04A2 – Boston Marine Industrial Park Tunnel (Accident location)	HDR Engineering, Inc. (10/90 – 1/93)	Kiewit / Perini / Atkinson / Cashman (JV) (5/92 – 3/95)	The C04A2 construction contract is the landside west tunnel approach to the Ted Williams Tunnel extending from the Ted Williams Tunnel immersed tube interface to approximately 200 feet west of D Street. The work consists of approximately 2,600 feet of I-90 cut-and-cover tunnel and one section of depressed open highway. The scope also includes a temporary ramp from I-90 to D Street.
I-90 Connector Tunnel Finishes			
C09B2 – I-90 Seaport Access Tunnel Finishes (Accident Location)	Gannett-Fleming (6/93 - 2/98)	Modern Continental Construction (11/98 – 7/02)	The C09B2 construction contract includes the interior tunnel finishes extending approximately 6,300 feet from the I-93 interchange, under the Fort Point Channel to the Ted Williams Tunnel. Major finish items in this contract include ceiling panels and structural support system, sidewall panels and structural support system, cast in place ceiling in gore areas, utility rooms and other areas, and roadway paving and striping.
Ted Williams Tunnel			
C05A1 – Immersed Tube Tunnel (East of the accident location)	Sverdrup Civil, Inc. (11/89 – 2/91)	J.F. White / Morrison – Knudson / Interbeton (9/91 – 6/94)	The C05A1 construction contract is the Ted Williams Immersed Tube Tunnel. The length of the immersed tube tunnel is approximately 3,850 feet.

Ted Williams Tunnel Finishes			
C05B1 – I-90 Tunnel Finishes (East of the accident location)	CPF / Domenech & Hicks, Inc. (11/90 – 7/92)	Walsh Construction Co. of Illinois (1/93 – 3/95)	The C05B1 construction contract includes the interior tunnel finishes that extends approximately 8,000 feet from the Boston Marine Industrial Park (BMIP) in South Boston (C05B1), under Boston Harbor (C05A1), to the Bird Island Flats (BIF) in East Boston (C07A1). The scope includes installation of ceiling panels and structural support system, light fixture support system, tile sidewall, polymer concrete walk finishes, utility room cross passages finishes, concrete floor and CMU walls finishes, roadway level exit doors and egress signing, roadway paving and striping.

5.1 VALUE ENGINEERING REPORT – TUNNEL FINISHES

A Value Engineering Report⁷ was prepared in anticipation of the tunnel finishes contracts for portions of the Central Artery / Tunnel (CA/T) Project that were not yet under construction. The total number of tunnel finishes contracts for the CA/T Project was three. The I-93 tunnel finishes contract combined five different tunnel finishes contracts into one construction contract, C17AA. Bechtel /Parsons Brinckerhoff (B/PB) decided that five different tunnel finishes contracts on I-93 would be too cumbersome for five different design consultants. An integrated design consultant was chosen to lead the I-93 tunnel finishes contract. Listed below are the tunnel finishes contract numbers and location.

- C05B1 – I-90 Ted Williams Tunnel Finishes Contract,
- C09B2 – I-90 Seaport Access Tunnel Finishes Contract, and
- C17AA – I-93 Tunnel Finishes Contract.

It was the intent of the Value Engineering Report to capture the lessons learned from the Ted Williams Tunnel finishes (C05B1) contract. The lessons learned would be a key part of contract documentation and administration for the remaining I-90 and I-93 tunnel finishes contracts.

⁷Commonwealth of Massachusetts Central Artery (I-93) / Tunnel (I-90) Project Value Engineering Report – Tunnel Finishes, March 1995, prepared for the Department of Public Works and Bechtel / Parsons Brinckerhoff by Olympic Associates Company.

The lessons learned from the Ted Williams Tunnel finishes (C05B1) contract were the following:

- Level of finish reduced,
- Ceiling panels simplified,
- Embedded anchorage system provided (in new tunnel areas),
- Labor intensive details, and
- Time-consuming to install.

One of the key recommendations from the Value Engineering Report was to study the costs associated with various ceiling options compared to the current design in the Ted Williams Tunnel. The various ceiling options were the following:

Option 0	Panel weight 30 lb/sf, cost \$50/sf approximate: porcelain enameled steel face precast panels on a 4' x 6' grid of hangers (current design)
Option I	Panel weight 35 lb/sf, cost \$44.69/sf: precast panels supported on a 4' x 6' grid (current design modified to remove porcelain enamel facing)
Option II	Panel weight 50 lb/sf, cost \$34.55/sf: precast lay-in panels supported on continuous inverted steel tees at 12' on center (tees are aligned with the lane edges)
Option III	Panel weight 50 lb/sf, cost \$46.23/sf: cast-in-place concrete supported on continuous inverted steel tees at 12' on center (tees are aligned with the lane edges)

The Value Engineering Report recommended the following option be selected⁸:

“B/PB recommends the selection of ceiling system Option II on the basis of maximum cost containment, for an estimated total projectwide savings of \$65 million.”

5.2 CONSTRUCTION OF I-90 CONNECTOR TUNNEL FINISHES (C09B2)

Modern Continental Company (MCC) was the low bidder for the I-90 connector tunnel finishes (C09B2) contract. Notice to proceed was granted on November 16, 1998. The installation of the ceiling hangars and ceiling modules in the I-90 eastbound tunnel generally proceeded from east to west in as linear fashion as originally planned due to the scheduled and actual sequence of turnovers from the I-90 tunnel contractors. The first I-90 eastbound tunnel

⁸Commonwealth of Massachusetts Central Artery (I-93) / Tunnel (I-90) Project Value Engineering Report – Tunnel Finishes, March 1995, prepared for the Department of Public Works and Bechtel / Parsons Brinckerhoff by Olympic Associates Company, 2.9 Tunnel Ceiling Alternatives, Summary and Recommendations, sheet 2 of 18.

section turned over to MCC was the 845 foot I-90 eastbound tunnel section from the east portal entrance at Station 72+65 to approximately Station 64+20.

Modern Continental's contract progress schedule was the following:

1. MCC constructed a tunnel finishes mock-up, in the nearby High Occupancy Vehicle (HOV) eastbound tunnel, from June 1999 through September 1999.
2. MCC installed the rod hanger assembly in the I-90 eastbound tunnel from Station 72+65 to Station 62+20 (in the area of the accident location) from October 5, 1999 to November 30, 1999.
3. MCC installed the ceiling modules and plenum in the I-90 eastbound tunnel from Station 72+65 to Station 62+20 (in the area of the accident location) from November 15, 1999 to March 13, 2000.
4. MCC installed the last I-90 eastbound ceiling module on July 10, 2002.
5. The I-90 eastbound connector tunnel was opened for traffic on January 18, 2003.

6. DESIGN OF CEILING MODULE

6.1 DESIGN OF ADHESIVE ANCHORS BY ENGINEERING CONSULTANT

Gannett Fleming, Inc., the section design consultant for the Central Artery (I-93) / Tunnel (I-90) (CA/T) Contract C09B2 (Tunnel Finishes) provided a brief summation of the design principles and relevant loading of the adhesive anchors by letter dated August 4, 2006 (Updated November 2, 2006) (See Attachment 1 – Memorandum to the National Transportation Safety Board from Gannett Fleming, Inc. dated August 4, 2006 (Updated November 2, 2006)). The concrete ceiling panels that fell on July 10, 2006 were located at approximately Station 71+00 in the eastbound travel lanes of the I-90 connector tunnel. The August 4, 2006 letter indicated the following:

“In response to your recent verbal inquiry for a brief summation of design principles and relevant loading the following is provided. A summary sketch (SK1) is also provided as well as AutoCAD drawings for the I-90 EB tunnel plan and typical sections in electronic format.

*CAT C09B2 – Station 70+50 EB – 72+40 EB
Summary of Modular Ceiling Design*

A modular ceiling panel arrangement was designed to separate the exhaust plenum from traveled roadway. Design Policy Memorandum (DPM) 107 dated 3/13/97 was the basis for the design of the modular ceiling panel/hanger system which was subsequently installed in Tunnel Finish Contract C09B2.

Mainline Contract C04A2 (east of Station 70+50) was sufficiently advanced precluding installation of embedded support beams (unistruts) which were subsequently installed in Mainline Contract C01A3 (west of Station 70+50). Accordingly, from Station 70+50 to 72+40, epoxy resin anchor systems were installed (in Contract C09B2) to support the modular ceiling panels.

The basis for the specified parameters for the epoxy resin system anchors is as follows:

- (a) The design service load per anchor was computed to be 2.6 kips (1.8 k DL, 0.8 k LL). This load is computed for the two anchor – hanger systems which are the governing hanger connections. This computation is based on a 5' x 12' tributary area that was consistent for hangers spacing (5') and panel span (12' maximum) within this region of the tunnel.
- (b) A proof test load of 125% of the design service load of 3.25 kips (i.e. 1.25×2.6 kips) was specified (Conformed Special Provisions 723.480, Section 3.07 G.1, dated 7/8/99).
- (c) The project criteria for suction load at 42 psf is in response to negative only exhaust pressures under extreme event considerations, i.e. fire, maintenance testing, etc. Applying the suction load in combination with dead load reduces the load per bolt to 0.5 kips (1.8 k DL – 1.3 k suction) during this extreme event.
- (d) A design epoxy transfer load of 4.0 kips (with an additionally specified factor of safety between ultimate strength and allowable strength of adhesive = 4.00) was specified (Conformed Special Provisions 723.480, Section 2.08 E.1.a, dated 7/8/99). The specified increase in service strength for the epoxy resin anchor system over the computed service design load is intended to supply an additional margin of safety should flexure cracking be present in the tunnel roof slab.
- (e) The contractor anchor adequacy submittal certified by PE seal dated December 17, 1999 indicated that the product to be supplied would attain an adhesive tensile allowable force of 6.35 kips/bolt (with a safety factor of 4 against ultimate).
- (f) The vertical tension force of 2.15 kips per anchor was specified for the 4-anchor arrangement. The hanger assembly consists of one vertical rod and two diagonal (tension only) rods, one of which will resist seismic lateral loads. The total vertical (dead/live) load of 5.2 kips (see (a) above) is carried by 4 anchors; therefore, the vertical (dead/live) load computed per anchor is $5.2 \text{ k} / 4$ which equals 1.3 kips/anchor. The vertical component of (tensile) seismic load is 3.2 kips. The connection is designed such that the action line of a diagonal passes through the anchor centerline at the WT /concrete

interface; therefore the vertical component of seismic load of 3.2 kips may be assumed to be carried by 2 anchors. Accordingly, the vertical seismic load computed per anchor is $3.2 \text{ k} / 2$ which equals 1.6 kips / anchor. Additionally, the Massachusetts Building Code (applicable for seismic considerations for this project) permits a 33% overstress under a seismic loading condition; therefore the total calculated load may be reduced 25% for comparison to service based allowable strength parameters. Thusly, the computed total vertical load per anchor is $(1.3 \text{ k} + 1.6 \text{ k}) \times 0.75 = 2.15 \text{ kips}$.

- (g) *Redundancy criteria:* Project redundancy issues were discussed in meetings with all tunnel finish section design consultants and the Management Consultant in the summer of 1997. The resulting design direction was to follow the same criteria for failure as developed for the C17AA contract (as indicated in the Purpose statement of Design Policy Memorandum No. 107, dated March 13, 1997). In C17AA, each modular ceiling unit was to be designed for "...a failure mode wherein the failure of any one support hanger assembly will not result in the collapse of that modular unit". This was referred to as the "one hanger out" scenario. GF analyzed the epoxy adhesive and concrete insert support systems for both DL + LL and DL + LL + EQ, eliminating the end hanger (which by inspection of the system was the critical location). For the "one hanger out" situations, it was confirmed that this condition provided safety factors within the guidelines of the January 23, 1998 Draft DPM for the respective components of the concrete insert (Unistrut) support systems. A meeting between GF and the Management Consultant (MC) was conducted on October 9, 1998 to address this issue with respect to the epoxy adhesive transfer load in the region from Sta. 70+50 to Sta. 72+40 (C04A2 region). At that time, GF was directed to utilize a reduction in the factor of safety for the epoxy transfer load (from 4.0) to 2.0 for the "one hanger out" scenario (within the 4A2 region). Two bolt connections for vertical hangers and four bolt connections for combined vertical and inclined hanger connections from Sta. 70+50 to Sta. 72+40 were evaluated for DL + LL and DL + LL + EQ for "one hanger out" scenarios for compliance with the October 9, 1998 understanding. The design for the epoxy adhesive anchors achieved the target factor of safety.
- (h) *In response to the inquiry as to what specific arrangement of "two hangers out" that would cause a failure.* This loading case was not evaluated explicitly in the design computations as this was not a project design criteria.
- (i) *Source of the "1.25 x working load" directive.* The 1.25 amplification factor was based on the anchor testing procedure specified for the Ted Williams Tunnel; this was furnished in the form of a sample spec. by the Management Consultant to Gannett Fleming via fax on 9/23/98. This fax transmitted was used as the basis for the development of project Special Provisions for C09B2. The project Special Provisions developed for C09B2 underwent many review / modification cycles as the design progressed with comment / correction from

the Management Consultant. For the bid documents, Gannett Fleming applied the 1.25 to the computed design load as discussed in (b) above and the numeric value of 3.25 kips was documented in the Conformed Special Provisions as outlined in (b) above.”

ORIGINAL DESIGN LOADS

1. Ceiling panel = 50 PSF
2. Support frame = 110 PLF
3. Live load = 25 PSF
4. Suction load = 42 PSF, UPLIFT
5. Panel width = 12'
6. Hanger spacing = 5'-0"
7. Tributary area = 12'X5'
8. Applied load per hanger = 5.2 kips
9. Applied load per bolt (2 bolt condition) = 2.6 kips
10. Test Load = 3.25 kips (2.6 x 1.25)
11. Per contractor's sealed calculations dated 12/17/99, adhesive tensile allowable per bolt = 6.35K"

Gannett Fleming, Inc. provided a sketch of a typical ceiling support framing section in the eastbound travel lanes of the I-90 connector tunnel as part of Contract C09B2 (Tunnel Finishes). Figure 5 shows a sketch of the typical ceiling support framing section.

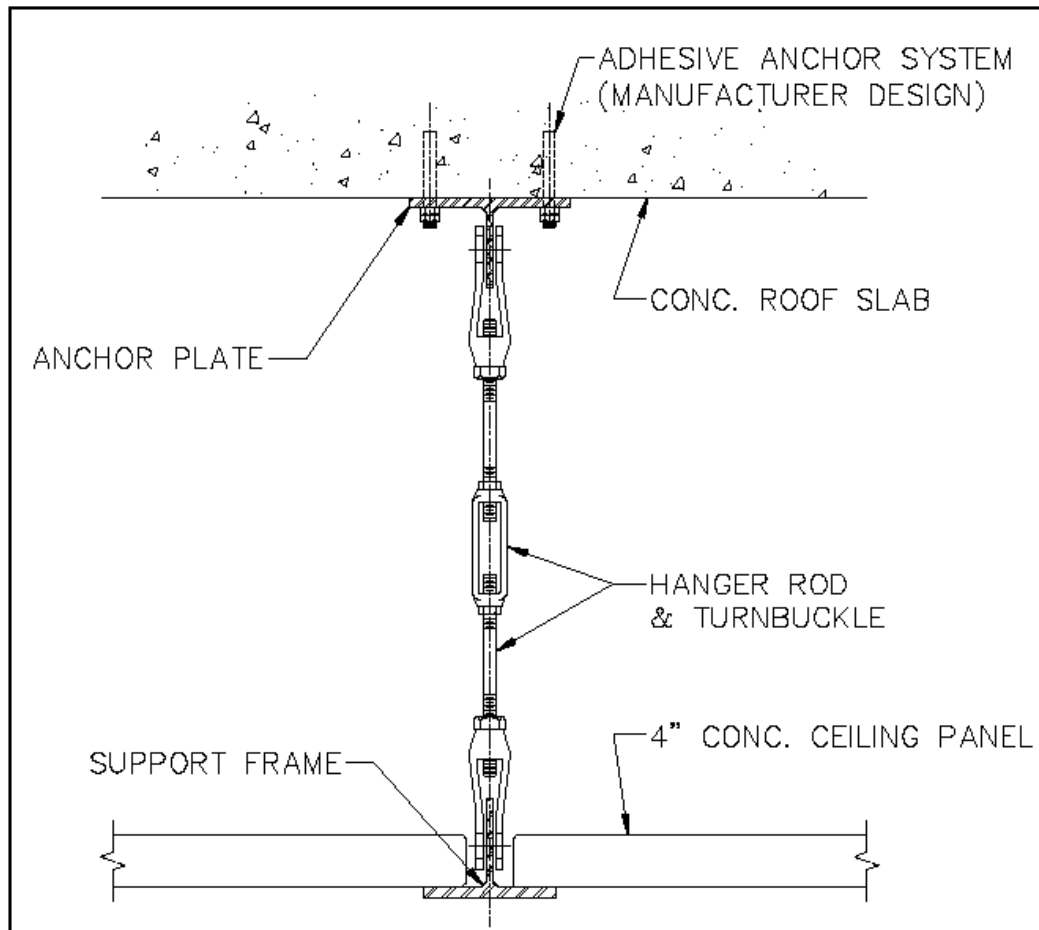


Figure 5 – Sketch of a typical ceiling support framing section in the eastbound travel lanes of the I-90 connector tunnel as part of Contract C09B2
(Courtesy of Gannett Fleming, Inc.)

6.2 CONTRACTOR ANCHOR ADEQUACY SUBMITTAL CERTIFIED BY PROFESSIONAL ENGINEER SEAL

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a contractor transmittal (submittal number 723.480-014 C) to Modern Continental Company (MCC) dated January 7, 2000 regarding the contractor's anchor adequacy submittal certified by Professional Engineer (PE) seal (See Attachment 2 – Contractor transmittal from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000). The contractor transmittal indicated the following:

“Work May Proceed Subject To The Changes Indicated, Resubmittal Not Required.

*Adhesive allowable loads (Power-Fast Epoxy Injection Gel)
 3/16 stainless steel rod allowables*

- a. Adhesive Tension Allowable = 6,350 pounds
- b. Adhesive Shear Allowable = 4,415 pounds
- c. Steel Rod Tension Allowable = 7,560 pounds
- d. Steel Rod Shear Allowable = 3,915 pounds

The allowables are based on the following minimum requirements:

Steel rod to be 3/16 stainless steel

Drill diameter = 3/4 inch

Concrete strength = 4,000 psi

Minimum anchor embedded length = 5 inches

Minimum anchor spacing (Tension or Shear Load) = $8d = 8 \times 0.625" = 5"$

Minimum edge distance (Tension Load) = $6d = 6 \times 0.625" = 3.75"$

Minimum edge distance (Shear Load) = $12d = 12 \times 0.625" = 7.5"$ Governs

Stamped and sealed by Massachusetts Registered Professional Engineer dated December 17, 1999"

6.3 DESIGN POLICY MEMORANDUM NO. 107 DATED MARCH 13, 1997

Gannett Fleming, Inc., the section design consultant, indicated redundancy⁹ issues were discussed in meetings with all tunnel finish section design consultants and the Management Consultant in the summer of 1997. The resulting design direction was to follow the same criteria for failure as developed for the C17AA contract as indicated in the purpose statement of Design Policy Memorandum No. 107, dated March 13, 1997 (See Attachment 3 – Design Policy Memorandum No. 107 Tunnel Finish Details dated March 13, 1997). The C17AA contract was the tunnel finishes contract for the I-93 northbound tunnel and I-93 southbound tunnel (excluding Kneeland Street to Congress Street southbound). In C17AA, each modular ceiling unit was to be designed for "...a failure mode wherein the failure of any one support hanger assembly will not result in the collapse of that modular unit". This was referred to as the "one hanger out" scenario.

The purpose statement of Design Policy Memorandum No. 107 Tunnel Finish Details¹⁰ indicated the following:

"This DPM provides advance information to tunnel finish SDCs regarding in-progress clarifications and changes to CA/T Directive Drawings resulting from the C17AA SDCs design development of the tunnel wall and ceiling details. It also outlines tunnel finish changes as a result of acceptance of certain Cost Containment proposals approved by MHD and FHWA at their Interface meeting

⁹Redundancy is a structural condition where there are more elements of support than are necessary for stability (i.e. failure of one hanger would not cause failure of the ceiling system because loads would be redistributed to adjacent hangers).

¹⁰Design Policy Memorandum No. 107 Tunnel Finish Details, from Massachusetts Highway Department Central Artery / Tunnel to DPM Manual Holders, March 13, 1997, page 1.

with B/PB. The Project intent to date has been that tunnel finish contracts would make use of the detailing by D017A as modified directives for other tunnel sections. Given the lead time before the completion of C17AA, it is important to transmit certain information now to avoid unnecessary, redundant or late work by other SDCs and to allow time to assess how the C17AA details may have to be modified to fit other sections' site specific requirements. It is not expected that CA/T Directive Drawings will be revised; the final detailing from C17AA will be transmitted at a later date by another DPM."

A general note appeared on the C17AA construction drawings¹¹ that referenced the "one hanger out" scenario:

"Each modular ceiling unit is designed for a failure mode wherein the failure of any one support hanger assembly will not result in the collapse of that modular unit."

6.4 FEDERAL HIGHWAY ADMINISTRATION'S (FHWA) FINITE ELEMENT ANALYSIS (FEA)

The Federal Highway Administration's (FHWA) Turner-Fairbank Highway Research Center developed a finite element analysis report to determine the loading conditions and the structural redundancy of the ceiling system (See Attachment 4 – Federal Highway Administration Turner-Fairbank Highway Research Center Finite Element Design Analysis Report). The finite element analysis included evaluating the ceiling system under the following three loading conditions:

- Dead Load 1 (DL1): The model was analyzed under gravity with all diagonal hanger rods removed from the system. This assumes that the diagonal hanger rods are not installed until after the concrete panels are in place.
- Dead Load 2 (DL2): The model was analyzed under gravity with all diagonal hanger rods present in the system. Note that the summation of anchor bolt loads is slightly higher in this configuration due to the weight of the additional hanger rods.
- Live Load (LL): A uniform pressure of 25 psf was applied in a downward vertical direction to the surface of the concrete panels. All diagonal hanger rods are present in the system for this load case.

Table 13 reflects the loading conditions on the ceiling system for the M1 panel support (the panel support that collapsed on July 10, 2006). With the diagonal hanger rods removed from the system (DL1), the maximum anchor bolt loads occur at the west end of the ceiling system. Based on field measurements, the hanger beams cantilever out six inches further on the west end compared to the east end. This created a slight asymmetry to the bolt force distribution.

¹¹C17AA Construction Drawings, I-93 Tunnel Finishes Northbound, Air-Plenum Ceiling System, Typical Notes and Details, Drawing No. H-114.

Table 13 - Bolt Force for Hanger Line EBD081-M1
(Courtesy of the Federal Highway Administration Turner-Fairbank Highway Research Center)

Location	Bolt ID	Bolt Force DL(1) Diagonals Out (lbs)	Bolt Force DL(2) Diagonals In (lbs)	Bolt Force LL Diagonals In (lbs)	Bolt Force DL(1)+LL (lbs)	Bolt Force DL(2)+LL (lbs)
1	N	1,589	1,086	466	2,055	1,552
	S	1,589	1,086	466	2,055	1,552
2-3-4	NE	911	1,387	584	1,494	1,971
	SE	911	1,387	584	1,494	1,971
	NW	911	1,375	572	1,483	1,947
	SW	911	1,375	572	1,483	1,947
5	N	1,586	1,280	539	2,125	1,819
	S	1,586	1,280	539	2,125	1,819
6	N	1,795	1,665	706	2,501	2,371
	S	1,795	1,665	706	2,501	2,371
7	N	1,597	1,471	620	2,217	2,091
	S	1,597	1,471	620	2,217	2,091
8	N	1,525	1,199	504	2,029	1,703
	S	1,525	1,199	504	2,029	1,703
9-10-11	NE	974	1,474	616	1,590	2,090
	SE	974	1,474	616	1,590	2,090
	NW	974	1,553	655	1,630	2,208
	SW	974	1,553	655	1,630	2,208
12	N	2,159	1,549	664	2,823	2,213
	S	2,159	1,549	664	2,823	2,213
SUM		28,044	28,078	11,853	39,897	39,931

Figure 6 shows the plan view of the ceiling system indicating the bolt labeling scheme used in Table 13.

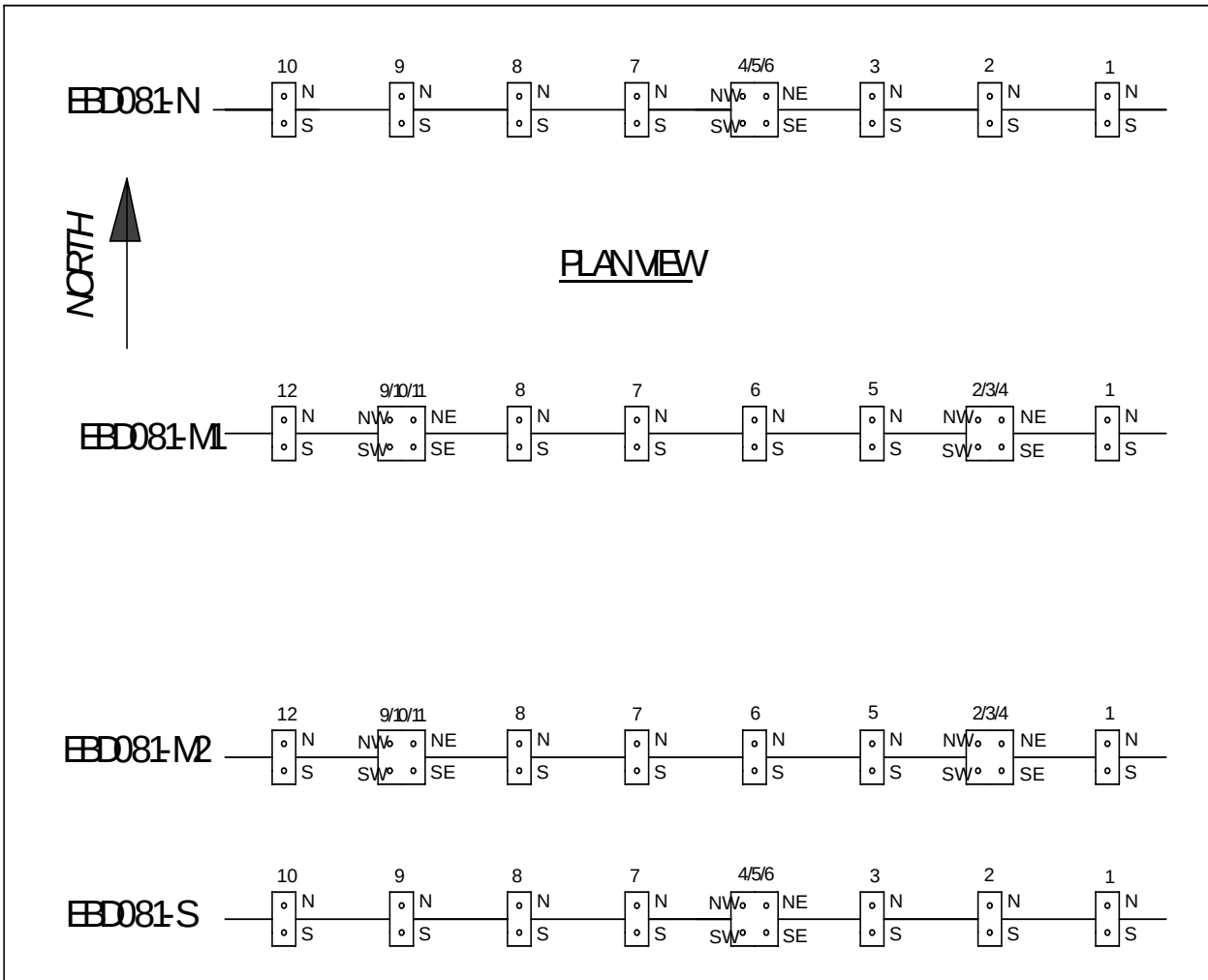


Figure 6 – Plan view of ceiling system showing the bolt labeling scheme
(Courtesy of the Federal Highway Administration Turner-Fairbank Highway Research Center)

The structural redundancy of the system was examined by FHWA by removing anchors from the model and repeating the computer analysis under both dead load and live load. Because the M1 row of anchors in the accident module had the highest proportion of load and were the anchors that failed in the accident, this was the only anchor row on which the redundancy analysis was performed.

In each failure scenario, all of the anchors (either two or four) were removed from a roof hanger plate to simulate a complete failure of one hanger assembly. The simultaneous removal of all the anchors for a hanger plate also provided a “worst-case” for redundancy evaluation.

FHWA researchers analyzed eight different “one-hanger-out” scenarios. Beginning at the east end of the M1 accident row, all of the anchors (either two or four) were removed sequentially from each roof hanger plate, and the results were analyzed. The analyses showed that failure of the second roof hanger from the west end of the row (hanger plate 9/10/11)

produced the highest load on the anchors at adjacent hanger plates. The maximum anchor load of 4,412 pounds was generated under the dead load plus live load condition and occurred at the two-anchor hanger at the west end of the row (hanger plate 12). This load is below the 6,350-pound maximum allowable load used in design considerations for the anchors.

Seven different “two-hanger-out” scenarios were investigated, with all of the anchors being removed sequentially from adjacent pairs of roof hanger plates. Each scenario, therefore, had either four or six anchors removed from the system. Results showed that a failure of the two hanger plates at the west end of the accident row (hanger plates 9/10/11 and 12) produced the highest load on the adjacent hangers. The maximum anchor load of 10,887 pounds was generated under the dead load plus live load condition and occurred at hanger plate 8. This load is well above the 6,350-pound maximum allowable load used in design considerations for the anchors, but still below the expected average short-term load capacity of 25,400 pounds.

Six different “three-hanger-out” scenarios were investigated, with all of the anchors being removed sequentially from three adjacent roof hanger plates. Each case therefore had either six or eight anchors removed from the system. Results showed that the most critical scenario involved a failure of the three westernmost hanger plates (hanger plates 8, 9/10/11, and 12). The subsequent maximum anchor load of 14,946 pounds was generated under the dead load plus live load condition and occurred at hanger plate 7. This load is also well above the 6,350-pound maximum allowable load used in design considerations for the anchors, but still below the expected average short-term load capacity of 25,400 pounds.

FHWA also determined the maximum deflection and maximum plate bending stress (Von Mises stress) for each of the “one-hanger-out”, “two-hanger-out”, and “three-hanger-out” scenarios. The results are shown in Attachment 4 – Federal Highway Administration Turner-Fairbank Highway Research Center Finite Element Design Analysis Report.

6.5 NTSB PHOTOS OF ADHESIVE ANCHORS BEFORE CEILING PANELS WERE REMOVED

NTSB investigators were on-scene immediately after the accident and took photos of the middle row of adhesive anchors in the westbound lanes of the I-90 connector tunnel. The photos show significant creep¹² and tensile movement of the adhesive anchors in the middle row of the westbound lanes (See STR Photo-01 through STR Photo-05) before the ceiling panels were removed.

6.6 MAXIMUM TORQUE VALUES

The manufacturer literature from Power’s Fasteners, Inc. for the adhesive anchoring system recommended maximum torque values and torque ranges for a given anchor size and drill diameter (See Attachment 5 – Manufacturer literature from Power’s Fasteners, Inc. on maximum

¹²Creep is a gradual, continuing, irreversible deformation due to a constant stress level below yield stress. Creep is a form of plastic deformation that occurs gradually at stress levels normally associated with elastic deformation. It is caused by molecular readjustments in a material under constant load (U.S. Department of Transportation, Federal Highway Administration, Safety Inspection of In-Service Bridges).

torque values and torque ranges for the adhesive anchoring system). The manufacturer literature recommended the following:

“51.5.6 Maximum Torque Values

A maximum torque value is listed for applications in which an adhesive anchor will be used to install a threaded rod. Although the application of torque is not necessary to achieve the published loads, in some cases it may be desirable to apply a clamping force to a fixture. The purpose of a maximum torque is to prevent over stressing of the adhesive bond. These values are based on testing in normal weight concrete at the standard embedment for capsule type adhesives and nine diameters of embedment for injection type adhesives. For specific applications, job site tests are recommended. As with mechanical anchors, preload relaxation should be expected due to creep within the concrete and in the adhesive. Refer to the section on Anchor Selection Guidelines for details.

Ultimate load capacities for installations using threaded anchor rod in concrete:

Anchor Size (inches)	Drill Diameter (inches)	Embedment Depth (inches)	Torque Range (ft-lbs.)	4,000 psi Concrete	
				Tension (lbs.)	Shear (lbs.)
5/8	3/4	5-5/8	50 - 90	30,175 lbs.	17,650 lbs.

The manufacturer literature from Power’s Fasteners, Inc. recommends maximum torque ranges of 50 to 90 foot-pounds for a 5/8-inch diameter adhesive anchor. A letter from Power’s Fasteners, Inc. dated October 29, 1999¹³ indicated that torque readings were obtained in the HOV ceiling mock up area of 120 foot-pounds. The letter indicated the following:

“Mr. Rice checked both the anchors that showed movement as well as anchors that had not moved and got torque readings of 120 foot-pounds. This torque value is greater than the published maximum torque range of a 5/8” diameter Power-Fast adhesive anchor (50-90 foot-pounds).”

6.7 PROOF TEST LOADS

A proof test load was mentioned in the special provisions of the I-90 Seaport Access Tunnel Finishes¹⁴ construction contract that indicated the following:

“Anchor shall be tension loaded to a service design load of 3.25 kips tension.”

Gannett Fleming, Inc. confirmed that the proof test load of 3.25 kips tension was a number derived from taking 125% of the design service load per anchor (i.e. 1.25 x 2.6 kips) for

¹³Letter from Power’s Fasteners, Inc. dated October 29, 1999 is attached to Attachment 12 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated November 8, 1999.

¹⁴Precast Concrete Ceiling System Section 723.480, I-90 Seaport Access Tunnel Finishes C09B2, April 22, 1998, Div. II Special Provisions, 2.08 (B1) Adhesive Type Ceiling Anchorage System, page 8 of 13.

the two anchor hanger system. The two anchor hanger system was the governing hanger connection.

Gannett Fleming, Inc. indicated the source of the “1.25 x working load” directive was based on the anchor testing procedure specified for the Ted Williams Tunnel that was furnished in the form of a sample specification by the Management Consultant to Gannett Fleming, Inc. via facsimile dated September 23, 1998 (See Attachment 6 – Facsimile Transmittal from Massachusetts Highway Department to Gannett Fleming, Inc. dated September 23, 1998 giving direction on proof test load for anchors). The facsimile was used as the basis for the development of project Special Provisions for the C09B2 construction contract. The facsimile indicated the following:

“Subject: Anchor Bolt Testing

Comments: Please incorporate/approve these changes based upon lessons learned in TWT.

3.07 INSTALLATION OF ADHESIVE TYPE CEILING ANCHORAGE SYSTEM

F. After chemical adhesive has cured, proof test all anchors to 125 percent of their design load per ASTM E 448, unconfined static tests for tension and shear. Record results of proof tests and submit to Engineer. Clearly mark and abandon anchors which fail test. Proof test procedure must be submitted to and reviewed by Engineer prior to installation.”

6.7.1 PROOF TEST LOADS FOR TED WILLIAMS TUNNEL

The proof test loads for the adhesive anchors in the Ted Williams Tunnel¹⁵ was the following:

“Each anchor was tested to 1.25 times its rated maximum working load, equal to 7,000 pounds for 3/4 inch diameter anchors.”

6.8 PROJECT PARTICIPANT RESPONSIBILITIES

Listed below was a partial list of project participant responsibilities for the I-90 Seaport Access Tunnel Finishes construction contract:

Massachusetts Turnpike Authority (MTA) / Massachusetts Highway Department (MHD)

- Ultimate decision maker for Central Artery / Tunnel (CA/T) project issues, including those affecting cost, claims and schedules,
- Assure adequate funding to implement the project,
- Procurement of professional consultants,

¹⁵Metropolitan Highway System Stem to Stern Safety Review – Phase I, Volume 3 – Tunnels – Finishes, November 15, 2006, page 16.

- Awards and executes all construction contracts,
- Oversight and management of Bechtel / Parsons Brinckerhoff (B/PB), and
- Operations and maintenance responsibility.

Bechtel / Parsons Brinckerhoff (B/PB)

- Preliminary design,
- Final design coordination and review,
- Construction coordination and monitoring,
- Preparing cost estimates and reporting to MHD/MTA on the project's overall cost and schedule, and
- Owner's authorized representatives for construction contract administration.

Gannett Fleming, Inc. (Section Design Consultant – SDC)

- The final design of each section will be the responsibility of a final design firm or team through the award of Section Design Consultant (SDC) contracts¹⁶,
- Develop and stamp detailed design sufficient for construction contractors to bid and build the various section contracts of the project,
- Provide construction phase services such as review of shop drawings, submittals, and responses to a construction contractor's Request for Information (RFI),
- The consultant is responsible for the correctness and completeness of the final design¹⁷,
- The consultant shall be responsible for the detailed design of the I-90 tunnel finish elements which include "Tunnel Ceiling Panels and Support System"¹⁸, and
- The consultant shall be responsible for structural design including design of structural supports for wall and ceiling finishes¹⁹.

Listed below was a partial list of Bechtel / Parsons Brinckerhoff's (B/PB) role in reviewing the Gannett Fleming, Inc. construction contract:

- Objective – design reviews are conducted with the primary objective of determining that the functional requirements of the CA/T project are provided within the restraints of approved project schedules and budgets²⁰,
- Detail of review – the review is not intended to lessen the responsibility of the SDC to review and check the calculations and design documents. Each SDC is responsible for completeness

¹⁶Work Program 15, 2.1.

¹⁷MHD/Gannett Fleming Contract, Attachment B "Scope of Services", paragraph 3.1 entitled "Review of Information".

¹⁸MHD/Gannett Fleming Contract, paragraph 3.2 entitled "Final Design", paragraph 3.2.1 entitled "Engineering and Architectural Design".

¹⁹MHD/Gannett Fleming Contract, paragraph 3.2 entitled "Final Design", paragraph 3.2.1 entitled "Engineering and Architectural Design".

²⁰Central Artery / Tunnel Project Procedure 411, Rev. 4, 4/92, paragraph 3.0 entitled "Instructions", paragraph 3.1.1 entitled "Objective".

and conformance to standards, codes, permit requirements, criteria, and other project requirements²¹,

- The reviewer will generally review documents for obvious errors, discrepancies, and conflicts,
- The reviewer will spot check to verify that the sizes and details of structural members are in agreement with the calculations submitted by the Section Design Consultant (SDC), and
- The reviewer will review random (nominally 3) structural elements or tunnel sections in each design contract, to independently verify the appropriateness and adequacy of the Section Design Consultant's (SDC) design.

6.9 DESIGN OF SUSPENDED CEILING PANELS IN OTHER TUNNELS THROUGHOUT THE COUNTRY

NTSB staff contacted nine tunnel authorities to determine, from a national perspective, the design of suspended ceiling panels used throughout the country. Information obtained from the nine tunnel authorities included the following:

- The extent of adhesive anchors used in other tunnels,
- Whether the adhesive anchors were engaged in shear or tension,
- The redundancy design used in other tunnels,
- The methods used to assure a continuous support of the end of the concrete panels to the tunnel walls,
- Inspection results used in other tunnels, and
- The dimension and weight of a typical ceiling panel.

Table 14 summarizes the design of suspended ceiling panels used in other tunnels throughout the country.

Table 14 – Design of suspended ceiling panels in other tunnels

Tunnel Authority	Tunnel Names	Design of Suspended Ceiling Panels
Massachusetts Turnpike Authority	Sumner Tunnel	The Sumner Tunnel ceiling system features hanger lines spaced nominally on 10-foot centers, with each line comprised of five vertical hangers secured to the concrete liner and supporting the 2-inch thick composite lightweight concrete enameled panels. Galvanized steel bent plates, or yokes, are secured to the concrete liner with a single ¾-inch diameter stainless steel post-installed adhesive anchor embedded 6-5/8 inches. Longitudinal and

²¹Central Artery / Tunnel Project Procedure 411, Rev. 4, 4/92, paragraph 3.0 entitled "Instructions", paragraph 3.1.2 entitled "Detail of Review".

		transverse braces are provided at 40-foot intervals along the length of the tunnel by diagonal pipe struts between the ceiling and the tunnel liner. Typical ceiling panel dimension and weight: Length = 11.25 feet (11 feet and 3 inches) Width = 3.25 feet (3 feet and 3 inches) Thickness = 0.1667 feet (2 inches) Weight = 550 pounds
	Callahan Tunnel	The configuration, components, and features of the Callahan Tunnel ceiling system are nearly identical to the Sumner Tunnel with the following variance: the ceiling panel used in the Callahan Tunnel is a 2-1/4-inch thick composite ceiling panel comprised of seven discrete layers (thin gauge enamel coated sheet metal, nominal 3/4-inch thick calcium silicate boards, and 1/4-inch thick cement fiber boards). Typical ceiling panel dimension and weight: Length = 11.25 feet (11 feet and 3 inches) Width = 3.25 feet (3 feet and 3 inches) Thickness = 0.1875 feet (2 and 1/4 inches) Weight = 550 pounds
	I-90 Ted Williams Tunnel	The Ted Williams Tunnel ceiling system consists of enameled sheet metal that fully encases, and is bonded to, a 2-inch thick core of lightweight concrete. The ceiling panels are typically bolted to the bottom flanges of longitudinal structural steel channel stringers, which in turn are suspended from the tunnel roof or concrete immersed tube tunnel liner by steel pipe struts. Hanger struts and diagonal braces are typically designed to act in tension and compression to resist design combinations of gravity, suction, and seismic loads. Each hanger and brace is connected to the cast-in-place concrete roof slab or tunnel liner by a U-shaped bent-plate yoke with two post-installed adhesive anchors. Typical ceiling panel dimension and weight: Length = 11 feet Width = 4 feet Thickness = 0.1667 feet (2 inches)

	I-90 Connector Tunnel	Weight = 1,200 pounds The I-90 Connector Tunnel ceiling system consists of 4-inch thick precast concrete panels that span simply in the transverse direction between longitudinal structural steel stringers, which in turn are suspended from the tunnel roof by steel rod hangers. The ceiling along each roadway comprises a series of structurally independent modules that are separated by transverse isolation joints spaced on nominal 40-foot centers. The ceiling panels were sized so that their weight would exceed the maximum design suction pressure, with the result that the hangers would remain in tension under all design gravity and suction loading conditions. Hanger connections to the ceiling typically consist of welded steel brackets that are bolted into embedded steel channel inserts cast into the concrete tunnel roof slab. The ceiling design specified post-installed adhesive anchors for the hanger connections in a short section near the east portal. Adhesive anchors were also installed at many other locations where installation of the hangers was hampered by misplacement of the channel inserts. Typical ceiling panel dimension and weight: Length = 12 feet Width = 8 feet Thickness = 0.3333 feet (4 inches) Weight = 4,800 pounds
	I-93 Central Artery Tunnel	The I-93 Central Artery Tunnel ceiling system is similar to those in the I-90 Connector Tunnel: 4-inch thick precast concrete roof panels span transversely between parallel longitudinal steel stringers in 40-foot long modules which are suspended by tension only hanger rods from the tunnel roof. In the typical I-93 Central Artery Tunnel ceilings, unlike those in the I-90 Connector Tunnel, the hangers and braces connect to a series of vertical steel gussets or “hanger plates”, each of which spans longitudinally across several connections to the steel roof girders or to cast-in-

		place connection inserts in the reinforced concrete roof slabs. Typical ceiling panel dimension and weight: Length = 12 feet Width = 8 feet Thickness = 0.3333 feet (4 inches) Weight = 4,800 pounds
MTA Bridges and Tunnels in New York	Brooklyn Battery Tunnel	Brooklyn Battery Tunnel has shear reinforcement extending horizontally into the tunnel concrete lining. The side placed expansion anchors are mechanically engaged with the concrete lining and were installed approximately 10 years ago. Each expansion anchor is placed in such a manner that it works both in shear and tension. There are no anchors working in pure tension. Typical ceiling slab dimension and weight: Length = 10.667 feet (10 feet and 8 inches) Width = 20.4167 feet (20 feet and 5 inches) Thickness = 0.4167 feet (5 inches) Weight = 12,380 pounds
	Queens Midtown Tunnel	The ceiling panels and adhesive anchors in the Queens Midtown Tunnel were completed in 2001 and use epoxy resin expansion anchors. All anchors in the Queens Midtown Tunnel, except very few, are engaged predominantly in shear action. There are no adhesive anchors acting in pure tension. Typical ceiling slab dimension and weight: Length = 10.667 feet (10 feet and 8 inches) Width = 20.5 feet (20 feet and 6 inches) Thickness = 0.4167 feet (5 inches) Weight = 12,479 pounds A consultant recently completed inspection of both tunnels checking safety of the suspension system. The preliminary report finds that both tunnels are safe for public and all suspension systems are fully engaged and functioning properly. See Figure 7 for suspension system diagram used in

		Brooklyn Battery and Queens Midtown Tunnels.
The Port Authority of New York and New Jersey Tunnels	Lincoln Tunnel	The center tube of the Lincoln Tunnel was built in 1937. The north tube was built in 1945 and the south tube was built in 1957. All 3 tubes have the same type of suspended ceiling construction. It is a reinforced concrete slab with the bottom layer of mesh extending approximately 2 feet over the end supports. The slab was cast in place with the cast-iron rings concrete liner providing a continuous end support. Typical ceiling slab dimension and weight: Length = 22 feet Width = 5 feet Thickness = 0.3958 feet (4 and 3/4 inches) Weight = 8,000 pounds
	Holland Tunnel	Both the south tube and the north tube were constructed in 1927. The original suspended ceilings (similar to the Lincoln Tunnel ceilings) were replaced in 1986. For a quick installation, every panel had inserts (3/4 inch diameter stainless steel threaded rods) that were connected to steel brackets, which are bolted to the cast-iron rings. For approximately two feet at the ends of the panel, the pre-cast slab had reinforcing bars extend into a keyway ledge carved in the ring liner. The top 2 1/2 inches of the slab was cast in place assuring a continuous support at the end of the panel. Typical ceiling slab dimension and weight: Length = 23 feet Width = 5 feet Thickness = 0.375 feet (4 and 1/2 inches) Weight = 8,000 pounds The past few biennial inspections found that the ceilings in the Lincoln and Holland Tunnels, including their supports, are still in good condition. Supports: Each panel is supported on a concrete shelf at either end and by three intermediate supports consisting of rolled steel sections. The two supports located

		towards the end of each panel are secured to the concrete tunnel liner with heavy-duty stainless steel mechanical expansion anchors. The center support hanger is secured with a positive bolted connection to the cast iron tunnel ring.
Detroit – Windsor Tunnel Authority in Michigan	Detroit – Windsor Tunnel	The Detroit – Windsor Tunnel ceiling slab forms the floor of the overhead exhaust duct. The ceiling consists of a reinforced concrete slab that is supported at mid-span by manganese bronze metal ceiling hangers. Two 6 inch by 3.5 inch by 3/8 inch steel angles are encased in a haunched section of the concrete slab and run longitudinally along the tunnel at its centerline. These angles, working compositely with the haunch, provide support for the concrete slab between the hangers. The ends of the reinforced concrete slab are extended into the ring liner assuring a continuous support. In 2003, a consultant determined that an evaluation of the ceiling hangers are adequate in their existing condition and configuration with respect to prevailing structural codes and standards for fire safety in roadway tunnels. Typical ceiling slab dimension and weight: Length = 10 feet Width = 20 feet Thickness = 0.4167 feet (5 inches) Weight = 11,000 pounds
South Jersey Transportation Authority	Atlantic City Expressway Tunnel	No suspended ceiling panels are used in the Atlantic City Expressway Tunnel. The roof system is comprised of an approximate 1.25 foot thick precast concrete roof beam and 2.75 foot thick poured-in-place reinforced concrete. Ventilation in the tunnel is accomplished through jet fans that are mounted in the corners of the tunnel section.
Maryland Transportation Authority	Fort McHenry Tunnel	The suspended ceiling panel system in the Fort McHenry Tunnel consisted of 2 inch thick concrete fill porcelain enameled panels. Each of the panels are supported by pipe hangers that are extended into the concrete ceiling with expansion anchors. The ends of the panels are supported with stringers and plates that connect to the tunnel wall with tapered concrete anchors and expander nuts.

		Typical ceiling panel dimension and weight: Length = 6.8125 feet (6 feet and 9-3/4 inches) Width = 2.3229 feet (2 feet and 3-7/8 inches) Thickness = 0.1667 feet (2 inches) Weight = 353 pounds
Chesapeake Bay Bridge and Tunnel District in Maryland	Chesapeake Bay Tunnel	The suspended ceiling system in the Chesapeake Bay Tunnel consisted of 5 inch thick cast-in-place concrete slabs. The slabs are supported by 1.5 inch diameter stainless steel hangers located in the middle spaced at 12 feet on center. The hangers extend into an upper socket embedded into the concrete ring ceiling that contains stainless steel strap anchors. The end of the slabs are supported by a ceiling slab key in which shear reinforcement (dowels #4) are extended from the slabs into the concrete tunnel wall assuring a continuous support. Typical ceiling slab dimension and weight: Length = 12 feet Width = 12 feet Thickness = 0.4167 feet (5 inches) Weight = 8,640 pounds
Pennsylvania Turnpike Commission	Allegheny and Tuscarora Tunnels	The suspended ceiling system in the Allegheny and Tuscarora Tunnels are two span continuous slabs corbelled into the tunnel walls with a redundant support system at mid-span consisting of three hanger rods per slab. The slabs were originally designed with a redundancy built in to be self supported in the event that all three hanger rods failed. Each of the hanger rods are attached to a separate threaded insert that is cast in place at the top of the arch. Typical ceiling slab dimension and weight: Length = 30 feet Width = 28.5 feet (28 feet and 6 inches) Thickness = 0.4167 feet (5 inches) Weight = 47,500 pounds Every 5 years, an in-depth inspection is performed at the tunnels. A walk through inspection of the tunnels structural components is conducted every 2 years. The 2 year inspection includes ceiling slabs,

		walls, and the arch. Hanger rods are sounded to ensure that they are intact and in tension. Every 6 months, the Maintenance Department conducts a walk through to check the hanger rods. The Commission has never had a ceiling slab failure. Ceiling slabs have deteriorated over time due to ground water infiltration, but this has only resulted in aggregate breaking off and falling onto the roadway below.
Virginia Department of Transportation	Midtown Tunnel	The Midtown Tunnel was built in 1962 in Portsmouth / Norfolk, Virginia under the Elizabeth River. The tunnel consists of one tube that carries two lanes of traffic. The ceiling anchorage system uses cast-in concrete anchors. Typical ceiling panel dimension and weight: Length = 6.5 feet (6 feet and 6 inches) Width = 2.5 feet (2 feet and 6 inches) Thickness = 0.1667 feet (2 inches) Weight = 407 pounds
	Hampton Roads Bridge Tunnel	The westbound tube of the Hampton Roads Bridge Tunnel was built in 1958 and the eastbound tube was built in 1974. The tubes were built in Hampton / Norfolk, Virginia under the Hampton Roads. The tunnel consists of two tubes that each carries two lanes of traffic. The ceiling anchorage system uses mechanical expansion anchors in the eastbound tube and cast-in concrete anchors in the westbound tube. Typical ceiling panel dimension and weight in eastbound tube: Length = 7.583 feet (7 feet and 7 inches) Width = 2.0 feet Thickness = 0.1667 feet (2 inches) Weight = 474 pounds Typical ceiling panel dimension and weight in westbound tube: Length = 12.75 feet (12 feet and 9 inches) Width = 1.0 feet Thickness = 0.1667 feet (2 inches)

	Big Walker Mountain Tunnel	Weight = 318 pounds The Big Walker Mountain Tunnel was built in 1972 in Bland County, Virginia. The tunnel consists of two tubes that each carries two lanes of traffic. The ceiling anchorage system uses steel anchorage assemblies cast into the concrete tunnel lining. Typical ceiling slab dimension and weight: Length = 38.5 feet (38 feet and 6 inches) Width = 31.0 feet Thickness = 0.5 feet (6 inches) Weight = 89,500 pounds East River Mountain Tunnel The East River Mountain Tunnel was built in 1974 in Bland County, Virginia. The tunnel consists of two tubes that each carries two lanes of traffic. The ceiling anchorage system uses steel anchorage assemblies cast into the concrete tunnel lining. Typical ceiling slab dimension and weight: Length = 38.5 feet (38 feet and 6 inches) Width = 31.0 feet Thickness = 0.5 feet (6 inches) Weight = 89,500 pounds Downtown Tunnel The westbound tube of the Downtown Tunnel was built in 1952 and the eastbound tube was built in 1986. The tubes were built in Portsmouth / Norfolk, Virginia under the Elizabeth River. The tunnel consists of two tubes that each carries two lanes of traffic. The ceiling anchorage system uses mechanical expansion anchors in the eastbound tube and cast-in concrete anchors in the westbound tube. Typical ceiling panel dimension and weight in eastbound tube: Length = 7.5 feet (7 feet and 6 inches) Width = 2.5 feet (2 feet and 6 inches) Thickness = 0.1667 feet (2 inches) Weight = 474 pounds
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		Typical ceiling panel dimension and weight in westbound tube: Length = 7.5 feet (7 feet and 6 inches) Width = 2.5 feet (2 feet and 6 inches) Thickness = 0.25 feet (3 inches) Weight = 703 pounds
	NAS Runway #29 Underpass	The NAS Runway #29 Underpass was built in 1977 in Norfolk, Virginia. The tunnel consists of two tubes that each carries two lanes of traffic. The tunnel has no suspended ceiling.
	Monitor-Merrimac Memorial Bridge Tunnel	The Monitor-Merrimac Memorial Bridge Tunnel was built in 1992 in Newport News / Suffolk, Virginia. The tunnel consists of two tubes that each carries two lanes of traffic. The tubes were fabricated as a unit rather than single tubes. The ceiling anchorage system uses mechanical expansion anchors for both tubes. Typical ceiling panel dimension and weight: Length = 7.25 feet (7 feet and 3 inches) Width = 3.0 feet Thickness = 0.1667 feet (2 inches) Weight = 544 pounds

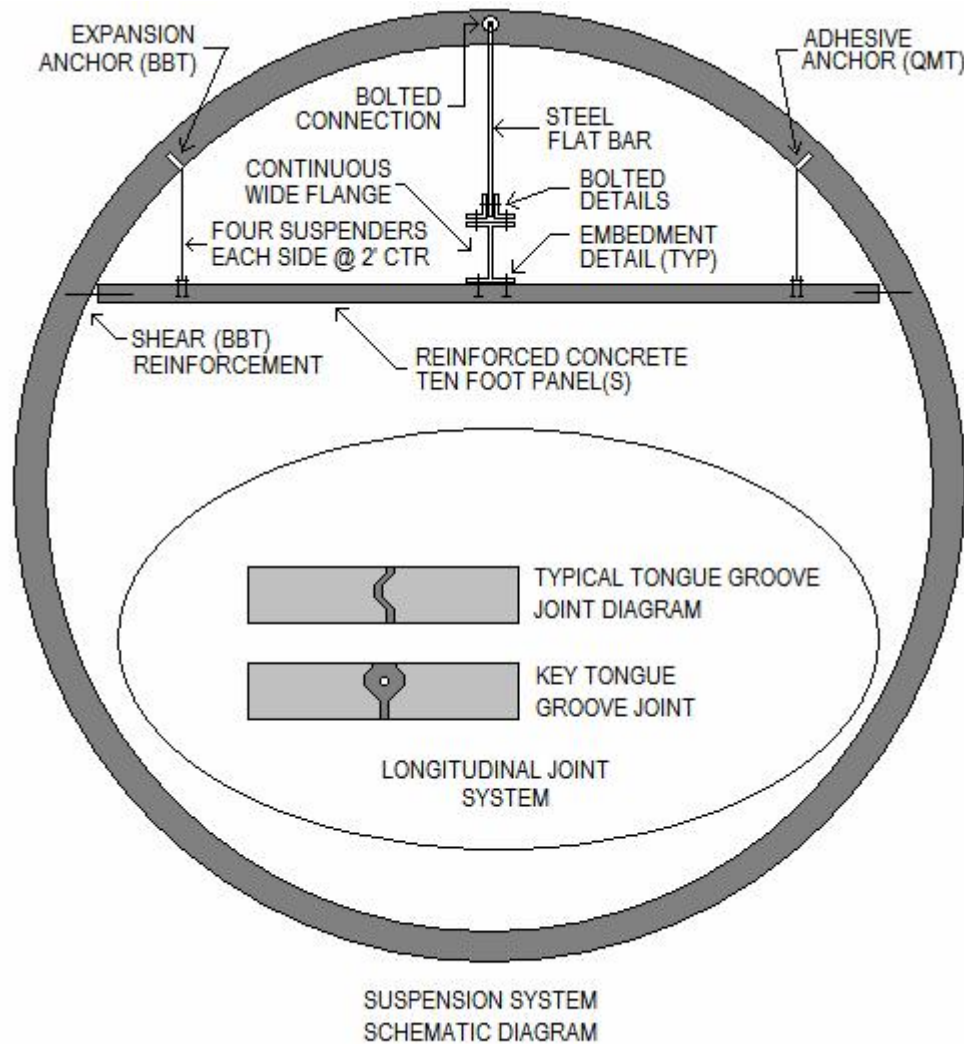


Figure 7 – Suspension system diagram used in Brooklyn Battery and Queens Midtown Tunnels

(Courtesy of the MTA Bridges and Tunnels in New York)

7. INSTALLATION PROCEDURE FOR THE CONCRETE CEILING PANELS

The installation procedure for the concrete ceiling panels was contained in Design Policy Memorandum No. 107, dated March 13, 1997 (See Attachment 3 – Design Policy Memorandum No. 107 Tunnel Finish Details dated March 13, 1997). The installation procedure was the following:

“Installation Procedure For The Modular Ceiling Support System:

1. *Install and properly align the hanger support plates and the associated hanger hardware from the tunnel roof girders.*

2. *The ten (10) panels are preassembled onto the two (2) panel support angles fastened together with the lifting angles by bolts at 24" oc.*
3. *Transport the modular units to the job site. Each modular unit is to be lifted by the eight (8) lifting lugs provided. All eight (8) points must be lifted simultaneously.*
4. *At the installation site, each unit is transferred onto a special vertical lift and lifted into position between a pair of hanger support plates. If not uniformly supported on the vertical lift, a minimum of eight (8) support points matching the lifting lugs are required.*
5. *Once into position, the single shear bolts are placed and tightened, connecting the panel support angles to the two (2) associated hanger support plates. The support system is flexible enough so that the units can be shifted slightly to accommodate construction tolerances.*
6. *After all lanes of ceiling panels have been installed, the tubular struts at the two (2) edges by the walls are installed. These struts are designed to take up any slack in the ceiling system in the transverse direction due to build in construction tolerances."*

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a contractor transmittal (submittal number 723.480-009 B) to Modern Continental Company (MCC) dated October 15, 1999 regarding the ceiling module erection procedure (See Attachment 7 – Contractor transmittal from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated October 15, 1999). The contractor transmittal indicated the following:

"Work May Proceed Subject To The Changes Indicated, Resubmittal Not Required.

Panels will be shipped to the jobsite on flatbed trailers with the panels set flat on the trailers. The panels will be trucked to the module fabrication area and directly installed from the trailer to the associated module. Wood timber beds will be set up within the module erection area to provide a level and clean working area for module assembly. These beds will be approx. 14 ft wide with the ability of the module lift equipment to span the bed, lift the module vertically, and carry the module to the lift trailer.

Structural steel elements shall be set into position on the wood timbers in the module fabrication area. The precast panels will be unloaded from the trailers using a material handler (i.e. Caterpillar 928 Loader) with a boom attachment, and placed into the support steel. Concrete Inserts shall be cast into the ceiling panels to facilitate handling of the panels in this manner. The precast panels will

then be bolted into position, and secured to the structural steel. The module will be inspected and prepared for hoisting onto the ceiling lift.

Upon completion of the module, the module will be lifted onto the trailer mounted ceiling installation lift using a mobile hoisting vehicle and then trucked to its intended location. The mobile hoisting vehicle (specifications attached) will be fabricated to work within the width and height constraints of the tunnel fabrication areas. The ceiling installation lift (specifications attached) is manufactured specifically for this purpose by Lift Systems, Inc. This lift has the ability to lift the heaviest module to the desired elevation and finely adjust the location in transverse and lateral directions to final location, along with the ability to tilt the module to follow the roadway profile. These lifts have proven to be effective on the existing Ted Williams Tunnel currently completed.

Upon lifting the module into place, the hanger assemblies will be fully connected to the module before the lift is lowered. Workers in manlifts will assist in the module installation and hanger attachment. Layout of the ceiling modules and tracking system will be included in alternate submittals.”

8. CENTRAL ARTERY / TUNNEL (CA/T) PROJECT VENTILATION SYSTEM DESCRIPTION AND OPERATION OVERVIEW

A description of the Central Artery / Tunnel (CA/T) Project ventilation system was provided by the Massachusetts Turnpike Authority. The following provides a description of the ventilation system:

“Tunnel ventilation systems servicing the Central Artery / Tunnel Project have been selectively designed to provide critical functioning during both normal traffic operation and emergency/fire conditions. The urban setting of the project also imposed significant demands on the tunnel ventilation system design and its allowable impact to the surrounding community. Sensitivity to land use and ambient environmental issues such as noise and air quality weighed heavily in determining the allowable size and locations of the necessary ventilation facilities. Full transverse, semi-transverse and longitudinal type ventilation systems were therefore utilized to meet both the functional demand of the various road tunnel configurations and the ambient environmental challenges in the most cost effective and efficient manner practical.

*From a tunnel ventilation perspective the Project is best defined as three distinct and separate road tunnel “systems”; the **Ted Williams Tunnel**, the **I-90 Tunnel Extension** and the **I-93 Central Artery Tunnel**. Each of these tunnel systems has been divided into multiple “ventilation zones”. Each ventilation zone is served by a dedicated and independently controlled set of fans that are housed in uniquely sited ventilation buildings. The project has a total of eight ventilation buildings and three ventilation shafts. In addition, in-tunnel mounted jet fans are utilized on ten different ramps. This concept allows for significant operational flexibility*

throughout the Project and provides the means for establishing the most efficient system operation under normal conditions and the most effective system operation in the case of a traffic incident or fire emergency. An inventory of ventilation buildings and fans is as follows:

Ted Williams Tunnel

Ventilation Building 7 (Logan Airport) – 24 Centrifugal Fans

Ventilation Building 6 (South Boston) – 10 Centrifugal Fans

I-90 Tunnel Extension

Ventilation Building 5 (South Boston) – 22 Centrifugal Fans

Ventilation Building 1 (Fort Point Channel) – 20 Centrifugal Fans

Two Ramps – 5 Jet Fans

I-93 Central Artery Tunnel

Ventilation Building 3 (Atlantic Avenue) – 23 Centrifugal Fans

Air Intake Shaft (Congress Street) – 2 Centrifugal Fans (supply only)

Ventilation Building 4 (Government Center) – 26 Centrifugal Fans

Ventilation Building 8 (North Station) – 2 Centrifugal Fans (supply only)

Summer Street Vent Shaft – 3 Vane Axial Fans

Essex Street Vent Shaft – 2 Vane Axial Fans

Beach Street Vent Shaft – 3 Vane Axial Fans

Eight Ramps – 29 Jet Fans

The CA/T Project's tunnel ventilation systems are controlled and monitored at the Massachusetts Turnpike Authority Operation and Control Center (OCC) in South Boston. From this facility, tunnel operators are assigned geographical areas of responsibility for oversight of all traffic management and support systems operation. Ventilation system control from this location may be either manual – allowing the operator to make specific adjustments – or automatic via a central computer based tunnel air quality algorithm or time-of-day histogram. In addition each of the tunnels ventilation systems may be controlled from the local ventilation facility.

During normal traffic operating conditions, the tunnel ventilation system is operated to maintain safe air quality and visibility within the tunnels. Carbon monoxide (CO) levels resulting from vehicle emissions are continuously monitored throughout all project tunnels. Real time values from each monitor are averaged by the OCC central computer system and reported on a per ventilation zone basis. Any exceedence of preset alert levels within a ventilation zone triggers an audible alarm to the operator and a banner display on the monitoring console provides specific data as to actual concentrations, trend and location. When controlled manually, the operator is then able to make any necessary adjustments to the ventilation zones in that particular tunnel area to restore safe air quality to the tunnel. Ventilation system adjustments might also be made during normal traffic operation when necessary to restore visibility that may be

degraded due to heavy fog or dust conditions. Other emissions generated within the tunnel, such as oxides of nitrogen and particulate matter, are monitored at points where they are released from the tunnels, i.e. exit portals and exhaust shafts, as these emissions may negatively impact local surrounding ambient conditions.

The tunnel ventilation systems have been pre-programmed to operate in the most effective modes for controlling smoke and heat in the case of a vehicle fire. The programming is based on system simulation modeling of severe fire conditions to determine the most effective way to achieve critical air velocity for all locations. When a fire occurs within any of the CA/T tunnels, the OCC operator would then bring up the ventilation system emergency operating matrix on his monitor and simply “click” on the column titled “fire location”. The central computer will then operate all necessary ventilation systems in their proper modes for securing as safe an environment as possible at the site of the fire.”

9. INSTALLATION GUIDELINES FOR THE ADHESIVE ANCHORING SYSTEM

9.1 POWER’S FASTENERS, INC. INSTALLATION GUIDELINES

The manufacturer literature from Power’s Fasteners, Inc. recommended the installation requirements for the adhesive anchoring system (See Attachment 8 – Manufacturer literature from Power’s Fasteners, Inc. on installation requirements for the adhesive anchoring system). The manufacturer literature recommended the following:

“52.8.1 Threaded Rods or Reinforcing Bars in Solid Base Materials

- 1. Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI Standard B212.15.*
- 2. Blow the hole clean with compressed air, brush the hole, and blow it clean again. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bars, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.*
- 3. Be sure to properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the hole approximately half way with epoxy starting from the rear of the hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets within the hole.*
- 4. Push the threaded rod or reinforcing bar into the hole while turning slightly to insure positive distribution of the epoxy. Be sure the rod is fully seated at the bottom of the hole and that some epoxy has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.*

5. Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.

Ultimate load capacities for installations using threaded anchor rod in concrete:

Anchor Size (inches)	Drill Diameter (inches)	Embedment Depth (inches)	4,000 psi Concrete	
			Tension (lbs.)	Shear (lbs.)
5/8	3/4	5-5/8	30,175 lbs.	17,650 lbs.

Allowable load capacities for installations using threaded anchor rod in concrete (applying a safety factor of 4 to the ultimate load capacity):

Anchor Size (inches)	Drill Diameter (inches)	Embedment Depth (inches)	4,000 psi Concrete
5/8	3/4	5-5/8	7,545 lbs.

9.2 AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO) GUIDELINES ON THE INSTALLATION, INSPECTION, AND TESTING OF EMBEDMENT ANCHORS

The American Association of State Highway and Transportation Officials (AASHTO) 2002 Standard Specifications for Highway Bridges²², 17th Edition, provided guidance on the installation, inspection, and testing of embedment anchors. AASHTO considered embedment anchors as cast-in-place, grouted, adhesive-bonded, expansion and undercut steel anchors. The AASHTO guidelines recommended the following:

“Section 29 EMBEDMENT ANCHORS

29.1 DESCRIPTION

This specification covers installation and field testing of cast-in-place, grouted, adhesive-bonded, expansion and undercut steel anchors.

29.2 PREQUALIFICATION

Conduct test for adhesive-bonded and other bonding compounds in accordance with ASTM E 1512 (Standard Test Methods for Testing Bond Performance of Adhesive-Bonded Anchors).

²²American Association of State Highway and Transportation Officials (AASHTO), Standard Specifications for Highway Bridges, 17th Edition, 2002, page 685.

29.4 CONSTRUCTION METHODS

If reinforcing bar is encountered during the drilling operation, move to a different location, or drill through the reinforcing steel using a diamond core bit as directed by the Engineer. Patch abandoned holes with an approved bonding material. Clean holes thoroughly as recommended by the manufacturer. Remove all loose dust and concrete particles from hole. Prepare bonding material and install anchors according to instructions provided by the manufacturer or approved by the Engineer. Embedded anchors which are improperly installed or which do not have the required strength shall be removed and replaced to the satisfaction of the Engineer at the Contractor's expense.

29.5 INSPECTION AND TESTING

Where specified, conduct sacrificial tests of the anchor system on the job site to ultimate loads to document the capability of the system to achieve pullout loads equaling the full minimum tensile value of the anchor employed. Test the anchor on fully cured concrete samples. Unless specified otherwise, test no fewer than three (3) anchors by ASTM E 488 methods. The Contractor may use any prequalified anchor systems meeting the above requirements.

Provide, without delay in progress, for an alternate system that will reach the designated pull-out requirement if the job site proofloading proves incapable of achieving minimum tensile values (or the designer's required load if too little concrete exists in which to develop full ductile loads).

After installing the curing of bonding material, torque each anchor system to values specified. If torque values are not specified, use values recommended by the manufacturer or provided by the Engineer."

10. LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED OCTOBER 7, 1999 INDICATING SMALL PERCENTAGE OF ADHESIVE ANCHORS IN THE HOV CEILING MOCKUP APPEAR TO SHOW SIGNS OF TENSILE MOVEMENT

Modern Continental Company (MCC) sent a letter to Bechtel / Parsons Brinckerhoff (B/PB) dated October 7, 1999 indicating a small percentage of the adhesive anchors in the HOV ceiling mockup appear to show signs of tensile movement, thus creating gaps in the ceiling hanger plate to ceiling interface (See Attachment 9 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated October 7, 1999). The letter indicated the following:

"As noted in our field inspection of October 7, 1999, a small percentage of the adhesive anchors in the HOV ceiling mockup appear to show signs of tensile movement, thus creating gaps in the ceiling hanger plate to CIP ceiling interface. The sequence of events leading to our notification is as follows:

- 1) *Ceiling module installed using proper securement prior to support removal in early August, '99. Inspection noted no movement subsequent to initial erection.*
- 2) *Small movement of anchor and attached plate (approx. 1/16") noted on September 9, 1999, by foreman installing plenums.*
- 3) *Periodic inspections by MCC revealed an increased movement to a worst case movement of 1/2" in early October of 1999, leading to notification.*

Whereas these anchors have been installed and tested in accordance with contract documents, MCC respectfully requests direction pertaining to any repair or mitigative measures required."

10.1 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED OCTOBER 12, 1999

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a response letter to Modern Continental Company (MCC) dated October 12, 1999 (See Attachment 10 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated October 12, 1999). The letter indicated the following:

"As discussed during our Engineering Issues meeting of October 8, 1999, and in response to the above referenced letter, the following was discussed.

We concur that the adhesive anchors were load tested in accordance with the design requirements, however we cannot confirm that the adhesive anchors are correctly loaded. A review of the ceiling system indicates that several turnbuckles are loose. This could induce excessive loading to the point that the anchors may be over stressed which may cause movement.

Other possibilities exist which could cause anchor bolt failure. During erection of the ceiling module the lifting system may have been released in such a manner to cause excessive stress to the anchors (too quickly) or prior to uniform loading of all turnbuckles. Failure to properly prepare & clean the drilled holes could cause a bond failure of the adhesive to the concrete over time.

The following action items were discussed and agreed to:

1. *MCC to contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors and obtain a manufacturer's recommendation.*

2. *MCC to confirm that all ceiling hangers are correctly loaded and have uniform load.*
3. *MCC to investigate with the manufacture the possibility to retest the anchors to design loading after the supports are properly adjusted.*
4. *B/PB will confirm that ceiling load calculations are correct.*

In summary, MCC shall confirm that the ceiling system is properly installed and all connection points are providing the correct support. Anchors that have moved since ceiling installation shall be abandoned and/or removed and replaced. When replacing the defective anchors, insure that proper measures are provided to support the ceiling so that other over stressed conditions are not created.”

10.2 LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED NOVEMBER 8, 1999

Modern Continental Company (MCC) sent a response letter to Bechtel / Parsons Brinckerhoff (B/PB) dated November 8, 1999 (See Attachment 11 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated November 8, 1999). The letter indicated the following:

“With regard to the adhesive anchor movement observed in Ramp HOV, Modern Continental Construction offers the following response.

In response to the observed anchor movement, and pursuant to discussions with the Engineer regarding this matter, Modern Continental Construction invited a technical representative from Powers-Rawl Fasteners to the site in order to assess the current condition of the anchors. This site visit was conducted on October 21, 1999. A summary of this site visit and recommendations for corrective action has been attached for your review.

In the above referenced correspondence, the Engineer cites improper ceiling module installation and anchor hole preparation procedures as potential causes for the anchor movement. Modern Continental and Powers-Rawl do not agree with these statements. If the anchors, were improperly installed (i.e. drill hole preparation, epoxy installation, etc.), it is likely that the anchors would have failed the proof test. As all of the anchors passed proof testing prior to module installation, it is not likely that anchor installation procedures were the cause of the anchor movement. Further, the ceiling modules were erected in accordance with the approved erection procedure, and in the presence of the Engineer. The assertion that the lifting devices were removed too quickly is unfounded.

Modern Continental Construction is currently conducting a review of the engineering associated with the adhesive anchors. Given the parameters associated with the ceiling design (approximate module weight, number of

anchors supporting the module, and tensile capacity of the anchor) and the applied factors of safety, it is improbable that the anchors were overstressed as a result of erection loads or the erection procedures. While the measured torque on some anchors exceeded recommended values, the manufacturer states that these anchors are functional, and should be subjected to a proof-load test for verification.

In summary, Modern Continental Construction confirms that all elements of the ceiling have been installed in accordance with the Contract Documents. Per the recommendations of Powers-Rawl, Modern Continental Construction will remove and replace the anchors which exhibited movement, and will proof test the anchors subjected to torque values higher than the manufacturer's recommended values in order to verify the anchor capacity. Modern Continental is currently evaluating available personnel and resources necessary to undertake this effort, and will inform the Project prior to the start of this operation.

However, in that the anchors have already passed the proof test and the root cause of the anchor failure has yet to be determined, Modern Continental considers this additional testing and support of this operation to be a change to the Contract. In accordance with Division I, Section 4.03, Modern Continental Construction reserves our rights to claim for additional costs associated with this change."

10.3 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED NOVEMBER 11, 1999

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a response letter to Modern Continental Company (MCC) dated November 11, 1999 (See Attachment 12 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated November 11, 1999). The letter indicated the following:

"Receipt of the above referenced letter in which Modern Continental Construction considers additional testing of adhesive anchors as a change to the Contract, is hereby acknowledged.

We do not concur with your position that all elements of the ceiling have been installed in accordance with the Contract. Whereas, adhesive anchors are over torqued and the ceiling hangers are not loaded uniformly. Until the anchors in question are removed and inspected can a determination be made of correct or incorrect installation techniques.

We concur with the recommendations in your letter from Power Fasteners that Modern Continental Construction will remove and replace anchors which

exhibited movement and the proof testing of the anchors subjected to torque values higher than the manufacturer's recommended values.

To summarize, the action items from our October 12, 1999 letter:

- 1. MCC to contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors and obtain a manufacturer's recommendation.*

Action has been completed, Power Fasteners has reviewed the installation and provided a recommended corrective action.

- 2. MCC to confirm that all ceiling hangers are correctly loaded and have uniform load.*

Open Item

- 3. MCC to investigate with the manufacturer the possibility to retest the anchors to design loading after the supports are properly adjusted.*

Action has been completed, Power Fasteners has reviewed the installation and provided a recommended corrective action to retest anchors that have not indicated movement.

- 4. B/PB will confirm that ceiling load calculations are correct.*

Action has been completed, B/PB has re-confirmed that design loads for the ceiling system are correctly stated.

Modern Continental Construction is requested to provide a schedule for the removal and replacement of the damaged anchors. B/PB requests to be present for the removal and retesting of the anchors in question. Prior to commencing new ceiling installations where adhesive anchors are used, confirm that no over torquing has occurred with previously installed ceiling plates.

Based on the above, the project has determined that there is no merit to MCC's claim for additional compensation and hereby inform you that this claim is denied."

10.4 LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED DECEMBER 2, 1999

Modern Continental Company (MCC) sent a response letter to Bechtel / Parsons Brinckerhoff (B/PB) dated December 2, 1999 (See Attachment 13 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated December 2, 1999). The letter indicated the following:

“In response to Deficiency Report No. 001 regarding Adhesive Anchor Failures in the area ceiling module HOV 079, attached you will find, based on the recommended actions, the corrective actions taken by MCC to resolve this issue.

DR-1 Adhesive Anchor Failures

In response to the above mentioned Deficiency Report the following corrective actions were / will be taken by MCC upon approval by B/PB.

- 1) On October 21, 1999 a Powers / Rawl Rep. was on site to investigate and determine to reason(s) for the failure of the anchors. Based on information gathered on site, which included a visual inspection of the anchors in question, a determination of failure could not be made.*
- 2) MCC submitted a letter / procedure from Powers Fasteners dated Nov. 30, 1999 to B/PB for approval (submittal no 723.480-016-A, also attached). Upon approval MCC will schedule a time to perform and complete this procedure.”*

10.5 CONTRACTOR TRANSMITTAL (SUBMITTAL NUMBER 723.480-016 A, ADHESIVE ANCHOR REPAIR PROCEDURE) FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED DECEMBER 27, 1999

Bechtel / Parsons Brinckerhoff (B/PB), the owner’s authorized representative, sent a contractor transmittal (submittal number 723.480-016 A) to Modern Continental Company (MCC) dated December 27, 1999 indicating the proposed fix for the failed anchor bolts, as submitted, is rejected (See Attachment 14 – Contractor transmittal from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated December 27, 1999). The contractor transmittal indicated the following:

“The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

Prior to beginning the repair process the contractor needs to submit documentation on the manufacturer of epoxy and type being used by Powers Fasteners. i.e., If it is a SIKA product what type is it?

The repair/replacement procedure for the failed anchors is as follows:

- 1. Remove the failed anchors without damaging the hole or adjacent concrete.*
- 2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole.*
- 3. Follow manufacturer’s procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening*

of the hole. Install a clean, new anchor and tension test the anchor using 125% of the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting allowable tension value is the adhesive bond strength. The 125% tension test value is 7900 lbs.

4. *Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.*

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. *Remove all of the anchors and follow the DR procedures to install new bolts, perform tension test on each bolt, reinstall hanger and torque bolts.*
2. *Retest all of the anchors using 125% of the governing Allowable Tension value, 7900 lbs."*

10.6 LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED DECEMBER 29, 1999

Modern Continental Company (MCC) sent a response letter to Bechtel / Parsons Brinckerhoff (B/PB) dated December 29, 1999 (See Attachment 15 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated December 29, 1999). The letter indicated the following:

"Modern Continental Construction is in receipt of the above referenced submittal regarding the adhesive anchor replacement procedure, and offers the following in response.

With regard to the replacement of the failed anchors, Modern Continental acknowledges that a clean drill hole is required prior to reinstallation of the new anchor, and if necessary will utilize a carbide drill bit to remove any remaining debris from the hole prior to installation.

With regard to the revised testing parameters for the failed anchors discussed in the above referenced submittal, Modern Continental Construction does not agree with the assessment of the Engineer. The referenced submittal comments indicates tensioning the anchor to 125% of the governing allowable tension value of the anchor, as indicated in the design catalog (approximately 7,900-lbs tension). This requirement is not consistent with the contractual requirement of proof testing to 3,250-lbs tension per ASTM E-448. Further, the submittal comments indicate a requirement to perform additional testing at the 7,900-lb tension on all of the installed anchors. Each of the installed anchors was successfully tested in

accordance with the contractual requirement of 3,250-lbs tension. Testing was performed in accordance with the approved testing procedure (reference submittal No. 723.480-011-B, dated 6/16/99). Test results have been forwarded to the Project, indicating compliance with the referenced contractual tensile strength requirements. The successful testing of each anchor substantiates compliance with the performance requirements for the adhesive anchor indicated in the contract.

The revised testing requirements indicated in the submittal review comments are not consistent with the testing requirements included in the Contract Specifications. Therefore, in accordance with Division I, Section 4.03, Modern Continental Construction respectfully requests that B/PB assign an Issue Number and subsequent Part I Modification prior to commencement of this additional work. Until such time, Modern Continental will track costs associated with this change as CP-022.”

10.7 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED JANUARY 7, 2000

Bechtel / Parsons Brinckerhoff (B/PB), the owner’s authorized representative, sent a response letter to Modern Continental Company (MCC) dated January 7, 2000 (See Attachment 16 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000). The letter indicated the following:

“Receipt of the above referenced letter, by which Modern Continental Construction Company (MCC) claims that the review comments as contained in Submittal Number 723.480-016 A are not consistent with the testing requirements included in the Contract Specifications, is hereby acknowledged.

The project has reviewed MCC’s claim and directs your attention to Division 1 of the General Requirements and Covenants of Construction Contracts, Section 5.04 wherein it states, “Regarding materials and installation techniques introduced by a Contractor Submittal that is accepted by the Engineer, in the event of differences among contract documents, the manufacturer’s written requirements and relevant codes and standards, in each instance the most stringent requirements shall apply. The Contractor shall perform the work thus specified at no increase in Contract Price or Contract Time”.

In the case of the ceiling anchor bolts, the bolts failed after the ceiling panels were installed. Examination of the one anchor that has been removed, indicated that the anchor bolt was improperly installed. There is evidence that the drill hole was not brushed clean, the application of a screen tube was not utilized and the anchor was not free of dirt, oil or foreign matter. Additionally the anchors were over torqued when the ceiling supports were installed.

The calculated allowable bond strength for a properly installed adhesive anchor system is 6350 lbs tension. MCC is required to demonstrate correct installation techniques of the adhesive anchor bolt system, therefore it is the Project's position that the re-testing of the bolts is required.

Division 1 of the General Requirements and Covenants of Construction Contracts, Sections 5.09 and 5.10 addresses inspection and removal of defective work. Retesting that confirms the manufacturer's calculated bond strength of 6350 lbs will be covered by a subsequent Contract Modification.

Based on the above the Project has determined that no additional compensation is due the contractor for the retesting of the installed anchor bolts. Therefore, MCC's claim is hereby denied."

10.8 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED JANUARY 7, 2000

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a letter to Modern Continental Company (MCC) dated January 7, 2000 (See Attachment 17 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000). The letter indicated the following:

"Attached for your action is the Project's corrective action for Deficiency Report Number 001 (DR-001), the failure of adhesive anchors in the HOV ceiling. Please incorporate these corrections into the work.

The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

The repair/replacement procedure for the failed anchors is as follows:

- 1. Remove the failed anchors without damaging the hole or adjacent concrete.*
- 2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole. Do not exceed manufacturer's tolerance for hole diameter. If hole diameter exceeds tolerance then change bits, hammer drill a larger diameter hole and install the next larger size anchor and maintain 5 inch hole depth.*
- 3. Follow manufacturer's procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening of the hole. Install a clean, new anchor and tension test the anchor using the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting Allowable Tension value is the adhesive bond strength, 6350 lbs.*

4. *Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.*

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. *Remove all of the anchors and follow the DR procedures to install new bolts. Perform tension test on each bolt using 6350 lbs, reinstall hanger and torque bolts.*
2. *Retest all of the anchors using the manufacturer's catalog and the governing Allowable Tension for adhesive bond strength – 6350 lbs.”*

10.9 LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED JANUARY 19, 2000

Modern Continental Company (MCC) sent a response letter to Bechtel / Parsons Brinckerhoff (B/PB) dated January 19, 2000 (See Attachment 18 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated January 19, 2000). The letter indicated the following:

“In response B/PB Letter No. 0498 regarding a denial of CP # 22, MCC does not agree with B/PB's position in regards to the determination of responsibility for re-testing previously tested and approved adhesive anchors.

B/PB quotes Division I of the contract stating that an approved submittal including manufacturer's data denoting an allowable adhesive bond strength of 6350 lbs now becomes a contract “requirement”, taking precedence over the original contract specification value of 3250 lbs. MCC asserts that the manufacturer's data represents merely information regarding the historic testing of the anchor and does not have any reference to a “requirement”. The bid upon requirement of this issue remains consistent with the contract specifications. B/PB's position would ultimately compel a contractor into submitting a lesser quality product, as no safety factor is allowed for the benefit of the contractor in providing a superior product.

Further, MCC's response to DR – 001 includes the manufacturer's requirements for remediating the pull out issues encountered on this project. An exercise which was directed by B/PB as part of the DR resolution. Re-testing of properly torques anchors or anchors to a higher strength criteria was not part of the manufacturers requirement of the contract documents.

MCC maintains that assuring any over torqued anchor meets the original contract specified load of 3250 lbs will be the responsibility of MCC. However, MCC will seek reimbursement of any additional testing of anchors torqued to manufacturer's recommendations or testing to a higher strength criteria beyond 3250 lbs consistent with previous notification of CP # 22.

MCC respectfully requests an FCN and associated Issue Number be provided in regard to this matter. If B/PB continues to maintain its original position requiring testing to 6350 lbs or re-testing previously tested, properly torqued anchors at the expense of MCC, MCC must request that this issue be immediately raised in accordance with the issue resolution model."

10.10 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED FEBRUARY 7, 2000

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a response letter to Modern Continental Company (MCC) dated February 7, 2000 (See Attachment 19 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 7, 2000). The letter indicated the following:

"Receipt of the above referenced letter, by which Modern Continental Construction Company (MCC) disagrees that the re-testing requirement of previously tested and approved adhesive anchors, is hereby acknowledged.

The calculated allowable bond strength for a properly installed adhesive anchor system is 6350 lbs tension. MCC is required to demonstrate correct installation techniques of the adhesive anchor bolt system, therefore the Project's position that the re-testing of the bolts is unchanged.

We disagree with Moderns position that testing to 6350 lbs is a contract change and as such, no FCN will be provided. We acknowledge your request to raise the issue in accordance with the issue resolution model. The issue will be added to our Senior Partners Meeting scheduled for Thursday, February 10, 2000 for development of a resolution plan."

10.11 ADHESIVE ANCHOR REPAIR PROCEDURE SUBMITTED BY MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED FEBRUARY 16, 2000

An adhesive anchor repair procedure was submitted by Modern Continental Company (MCC) to Bechtel / Parsons Brinckerhoff (B/PB) dated February 16, 2000 (See Attachment 20 – Adhesive Anchor Repair Procedure submitted by Modern Continental Company to Bechtel / Parsons Brinckerhoff dated February 16, 2000). The adhesive anchor repair procedure indicated the following:

“Work May Proceed Subject To The Changes Indicated, Resubmittal Not Required.

At this time, the proposed revised anchor testing procedure is applicable to the adhesive anchors installed in the mock-up area in Ramp HOV.

Per the Direction received on the previous submittal and Letter No. C09B2 – 0498, January 7, 2000 the proposed adhesive anchor repair procedure shall be modified as follows:

- 1. The failed anchor shall be removed from the concrete without damage to drill hole or adjacent concrete.*
- 2. A hammer drill with a carbide bit shall be utilized to re-drill / ream the hole to create a surface free of debris from the removed anchor. All foreign matter will be removed from the hole.*
- 3. A new adhesive anchor shall be installed per the approved submittal (Submittal 723.480-008 D, Section 3, entitled “Installation Procedure & Quality Control Program – Adhesive Anchoring System”).*

The new anchor shall be tested in accordance with the directive received in Letter No C09B2 – 0498, dated January 7, 2000. The revised loading requirement for proof testing shall be at 6,350 lb. tension.

- 4. The hanger attachment plate will be reinstalled and bolted to the adhesive anchors. Anchor bolts shall be torqued in accordance with Manufacturer’s recommendations.*

After compliance of the revised testing procedure in the Ramp HOV Mock-Up Area, MCC shall provide test results to the Engineer for review, and await further direction.

See attached for additional comments.

Submittal Review Comments from Field Engineering Staff C09B2

The proposed repair procedure seems to be written pertaining to “the” failed anchor, as if there is only one. It should pertain to “any” failed anchors.

Per Step 2 of Procedure: “All foreign matter will be removed from the hole”. Procedure for removing foreign matter should include 1.) blowing out the hole with compressed air, 2.) brushing out the hole (preferably with a steel or bronze wire brush – manufacturer recommends nylon brush), 3.) blowing out the hole again until no dust remains.”

10.12 INTERNAL MEMORANDUM FROM STAFF OF BECHTEL / PARSONS
BRINCKERHOFF DATED AUGUST 31, 2000 RECOMMENDING
DISPOSITION OF DEFICIENCY REPORT (DR) NO. 1 BE CLOSED

An internal memorandum from staff of Bechtel / Parsons Brinckerhoff (B/PB) dated August 31, 2000 recommended disposition of Deficiency Report (DR) No.1 be closed (See Attachment 21 – Internal Memorandum from Staff of Bechtel / Parsons Brinckerhoff dated August 31, 2000). The internal memorandum indicated the following:

“Recommended disposition for DR #1

Subsequent to the writing of DR #1, all epoxy anchors in the subject area were retested at the higher load of 6,350 lb. There were a total of 19 anchors that failed out of the 187 that were retested at the higher load of 6,350 lb vs. the initial contractly required load of 3,250. Refer to the break out below that gives the locations of the failures.

Module HOV078: 20 anchors retested, 3 failed. [located in mock-up]

Module HOV079: 55 anchors retested, 14 failed. [located in mock-up]

Module HOV080: 56 anchors retested, 2 failed.

Module HOV079: 56 anchors retested, 0 failed.

Each of the failed anchors was replaced and retested. Of these only one failed upon retesting and it was then replaced and retested again and passed. Hence, all existing anchors in the subject area have been tested and replaced as necessary and retested until passing at the higher load. Copies of the test reports from Conam Inspection, Inc. are attached.

The “subject area” is comprised of all epoxy anchors that were installed by C09B2 in the ceiling of HOV in the C04A2 Contract area, which comprises the area where ceiling modules designated HOV078, HOV079, HOV080, HOV081 are located.

Observation: There were only two failures in the modules located outside of the ceiling mock-up area. This could be the result of better installation techniques being used by the contractor as gleaned from going through the “learning curve” in the mock-up area. These better installation techniques include: (1) cleaning holes out better and not using first epoxy out of the gun during the installation of the epoxy anchors and (2) not over torquing the nuts when the ceiling plates are installed over the anchors.

It is my recommendation that this DR #1 be closed.”

10.13 LETTER FROM MODERN CONTINENTAL COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED SEPTEMBER 18, 2000 REQUESTING A CONTRACT MODIFICATION FOR RETESTING OF ADHESIVE ANCHORS TO 6,350 POUNDS

Modern Continental Company (MCC) sent a letter to Bechtel / Parsons Brinckerhoff (B/PB) dated September 18, 2000 requesting a contract modification be issued for reimbursement of the additional work associated with retesting the adhesive anchors to 6,350 pounds (See Attachment 22 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated September 18, 2000). The letter indicated the following:

“Per the above reference correspondences and conversations between Modern Continental and the Project, MCC has completed re-testing all ceiling adhesive anchors in the C04A2 area. The anchors were originally tested to 3,250 lbs tension, per the contract, and then re-tested to 6350 lbs, as directed by the Project. Throughout the duration of the work, time was documented on Time & Materials (T&M) slips.

In accordance with Division 1, Section 4.04 of the General Requirements and Covenants of Construction Contracts, Modern Continental Construction hereby submits a detailed cost proposal for Contract Proposal (CP) No. 22. The total cost associated with this change, is Seventeen Thousand One Hundred Two Dollars and Sixty-One Cents (\$17,102.61). MCC respectfully requests that a Contract Modification be issued as reimbursement for the additional work associated with this change.”

10.14 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED JANUARY 26, 2001 INDICATING CORRECTIVE ACTION HAD BEEN VERIFIED AND DEFICIENCY REPORT NO. 1 WAS CLOSED

Bechtel / Parsons Brinckerhoff (B/PB), the owner’s authorized representative, sent a letter to Modern Continental Company (MCC) dated January 26, 2001 indicating corrective action had been verified and Deficiency Report No. 1 was closed (See Attachment 23 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 26, 2001). The letter indicated the following:

“Attached for your use is the final disposition of DR No. 1, Adhesive Anchor Failures. DR No. 1 is closed.

CORRECTIVE ACTION VERIFIED:

All HOV A/B retested to new criteria failures have been replaced and retested.

Signed by Resident Engineer dated 1/26/01.”

10.15 LETTER FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED MARCH 15, 2001 ENCLOSING A FULLY EXECUTED ORIGINAL OF THE CONTRACT MODIFICATION

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a letter to Modern Continental Company (MCC) dated March 15, 2001 enclosing a fully executed original of the contract modification (See Attachment 24 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated March 15, 2001). The letter indicated the following:

“Enclosed, for your records, is a fully executed original of the above referenced Contract Modification.

Description:

Provide compensation to the contractor for re-testing the ceiling adhesive anchor bolts in the HOV Ramp of the C04A2 area.

Justification:

Due to several failures of ceiling anchor bolts in the HOV, the Project determined that all of the bolts that were installed had to be re-tested at 6350 lbs tension. The 6350 lbs represents the manufacturer's allowable bond strength for a properly installed adhesive anchor. The Contractor complied and tested all the anchor bolts but made claim for the re-testing costs as in their opinion, the Contract calls for 3250 lbs tension.

This issue was scheduled for hearing at Senior Partners since both parties could not initially come to agreement. However, in an effort to resolve this issue and not go to Senior Partners, since the total commercial value was minimal, an agreement was reached. The Project agreed to pay the re-testing costs for any anchor, which passed testing at 6350 lbs, and, to split the cost for anchors which were tested and passed at the 3250 level but failed to the 6350 level.

001.099 RETESTING OF CEILING ANCHORS IN HOV

\$16,396.50”

11. **DECEMBER 1998 REPORT FROM THE COMMONWEALTH OF MASSACHUSETTS OFFICE OF THE INSPECTOR GENERAL REGARDING USE OF ANCHOR BOLTS ON THE C05B1 TUNNEL FINISHES CONTRACT**

The Commonwealth of Massachusetts Office of the Inspector General released a report dated December 1998 that reviewed the Central Artery / Tunnel (CA/T) project's use of anchor bolts on the C05B1 Tunnel Finishes Contract (See Attachment 25 – The Commonwealth of Massachusetts Office of the Inspector General, A Review of the Central Artery / Tunnel

Project's Use of Anchor Bolts on the C05B1 Tunnel Finishes Contract). The contract included the installation of wall tiles, a ceiling, a roadway, signage, lighting, and finish details. The Office of the Inspector General focused on two change orders relating to the installation of the ceiling support system in the Ted Williams Tunnel. The report documents examples of poor design coordination and unclear contract specifications that led to the payment of \$850,000 for two no-bid change orders. The recommendations in the report aim at proactively assisting project managers to contain costs as they complete design and award the remaining tunnel finishes contracts, which are valued at more than \$200 million.

The reports findings and recommendations include the following:

"The \$10.8 billion Central Artery / Tunnel (CA/T) Project involves constructing a new tunnel across Boston Harbor, placing the Central Artery underground, and constructing a new Charles River crossing. Since the early 1980's, the Massachusetts Highway Department (MassHighway) has had the responsibility for planning and overseeing the Project for the Commonwealth. In 1985, MassHighway hired the joint venture of Bechtel/Parsons Brinckerhoff (B/PB) to manage the design, construction, and day-to-day administration of the CA/T Project. B/PB prepares preliminary design documents, manages final design contracts and construction, and provides administrative and technical support.

As construction manager, B/PB managed the C05B1 – I-90 Tunnel Finishes contract. This contract was for the installation of wall tiles, a ceiling, lighting, signage, and other items needed to make the tunnel usable.

This Office's review disclosed the following:

Finding One: *Poor design specifications created construction difficulties that cost almost \$800,000 to resolve.*

Finding Two: *The Section Design Consultant prepared poor design specifications for anchor bolt installation.*

Finding Three: *The Section Design Consultant prepared unclear testing procedures.*

Finding Four: *The Project paid the contractor to test improperly installed anchor bolts.*

Finding Five: *The Project issued a change order to compensate the contractor for poor subcontractor performance.*

Finding Six: *The Project did not consult with the tunnel designers before allowing the contractor to drill through steel reinforcement in the tunnel roof.*

This Office makes the following recommendations to assist MassHighway – through B/PB, where appropriate – in reducing future contract costs:

- 1) Direct B/PB to develop clear and complete specifications for construction contracts.*
- 2) Ensure better design coordination between contracts.*
- 3) Hold contractors accountable for shoddy work and poor planning.*
- 4) Ensure that Section Design Consultants hired by the Project to complete designs review and approve design and specification changes.*
- 5) Take cost recovery action when costly errors and omissions are discovered.*
- 6) Direct B/PB to ensure that contractors proceed with work only under approved procedures.”*

12. PROPOSAL BY MODERN CONTINENTAL COMPANY (MCC) FOR A LIGHTER CEILING SYSTEM

Modern Continental Company (MCC) sent a letter to Bechtel / Parsons Brinckerhoff (B/PB) dated January 13, 1999 requesting consideration of a Celline Metal Panel Structural Ceiling System as a Value Engineering Cost Proposal in lieu of the concrete ceiling panels (See Attachment 26 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated January 13, 1999). The cost savings for the Celline Metal Ceiling System was approximately \$4,000,000 compared to the concrete ceiling panels. The Celline Metal Ceiling System weighed less than 10 pounds per square foot and the concrete ceiling system weighed approximately 50 pounds per square foot. Listed below is an excerpt from the January 13, 1999 letter.

“.....In accordance with Division I Section 8.16, MCC respectfully requests consideration of a Celline Metal Panel Structural Ceiling System manufactured by Environmental Interiors, Inc. as a Value Engineering Cost Proposal in lieu of the precast roof structure, cast-in-place decking, and corrugated decking plenum System as indicated on the above referenced project. This System can be manufactured to suit engineer and owner requirements in either epoxy painted galvanized steel or stainless steel and has been previously used on several Federal projects.....”

Attached to the January 13, 1999 letter was a letter from Environmental Interiors, Inc. to Modern Continental Company (MCC) dated December 8, 1998. The letter indicated the weight of the Celline Metal Ceiling System:

“The entire Celline system (including optional full backer sheet) weighs less than 10 pounds per square foot.”

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a response letter to Modern Continental Company (MCC) dated February 24, 1999 indicating the Value Engineering Cost Proposal was not acceptable (See Attachment 27 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 24, 1999). Listed below is an excerpt from the February 24, 1999 letter.

“.....The proposed VECP No. 1 was evaluated by Project personnel representing appropriate disciplines who reached the following conclusion. The proposed C09B2 VECP No. 1, as presented, is not acceptable.

The following items would have to be incorporated, as a minimum, for “contingent” acceptance of the VECP. Final acceptance of the VECP and, therefore, permission to use this system in the C09B2 contract, shall be withheld pending; 1.) Review and acceptance of the vendor's revised design and their structural calculations by the Project; 2.) Review and acceptance of prototype panel testing results; and 3.) Project assessment of the initial system cost and its life cycle cost.....”

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a letter to Modern Continental Company (MCC) dated March 25, 1999 indicating the Value Engineering Cost Proposal should not be pursued further and was considered not viable to implementation for the C09B2 project (See Attachment 28 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated March 25, 1999). The letter indicated the following:

“Confirming our conversation today, we concur that VECP-001 should not be pursued further and is considered not viable to implementation for the C09B2 I-90 Seaport Tunnel Finishes Project. Considering the current status of your material purchase orders, the precasters expended engineering and buy-out costs, and unresolved engineering issues we agree that a substantial cost and/or schedule savings for this project do not appear feasible.

We appreciate your efforts to bring forth “Value Engineering” for consideration by the Project and encourage you to continue consider opportunities in the future.”

Environmental Interiors, Inc. withdrew the proposal for the Celline Metal Panel Ceiling System (See Attachment 29 – Letter from Environmental Interiors, Inc. to Modern Continental Company (MCC) dated March 9, 1999). Listed below is an excerpt from the March 9, 1999 letter.

“.....Please be advised that we are withdrawing our various proposals for the use of our Celline ceiling system in lieu of the precast concrete ceiling panels and support systems specified for this project. Unfortunately, our extended efforts continue to be met with significant resistance from the project design team. We

have been unable to overcome this resistance, primarily due to the Authority's unwillingness (or inability) to provide specific numerical standards, ASTM designations etc. for our compliance. Alternatively, we have been faced with vague and repeated "concerns" about vibration, corrosion and life cycle cost without further specific performance criteria. Our proposals have complied with every specified performance criteria from the contract documents and we have provided evidence that our G235 galvanized and epoxy coated Celline system would provide the Authority with its desired (not specified) 75 year life cycle. Despite our effort, the Authority and its design team have put us through several iterations of our VECP proposal, each resulting in cost increases and all without basis in hard factual data and without relation to the contract documents....."

13. CONTRACTOR TRANSMITTAL FOR USE OF SCREEN TUBES

Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, sent a contractor transmittal (submittal number 723.480-008 A) to Modern Continental Company (MCC) dated May 27, 1999 regarding the installation procedure of the adhesive anchoring system and the use of screen tubes (See Attachment 30 – Contractor transmittal from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated May 27, 1999). Listed below are excerpts from the contractor transmittal:

".....Revise and Resubmit. Work May Not Proceed.

Section 2 – *Installation Procedure & Quality Control Program*
Adhesive Anchoring System

5. *On floor applications, the drill hole shall be filled approximately one half full with the epoxy adhesive, starting from the rear of the hole.*

On wall and ceiling applications, a screen tube shall be utilized to ensure proper application of the epoxy. Upon confirming the fit of the anchor in the drilled hole, the screen tube shall be filled with epoxy, starting from the rear of the screen tube. The screen tube shall be inserted into the drill hole, and fully seated at the required embedment depth. The hole shall be plugged with an anchor seal plug....."

The manufacturer literature from Power's Fasteners, Inc. attached to the May 27, 1999 contractor transmittal indicated that the screen tubes should be used for threaded rods in hollow base materials. The manufacturer literature does not mention using screen tubes for threaded rods or reinforcing bars in solid base materials.

A second contractor transmittal (submittal number 723.480-008 C) was sent by Bechtel / Parsons Brinckerhoff (B/PB) to Modern Continental Company (MCC) dated November 19, 1999 indicating the use of screen tubes as a part of the installation procedure of the adhesive anchoring system.

A third contractor transmittal (submittal number 723.480-008 D) was sent by Bechtel / Parsons Brinckerhoff (B/PB) to Modern Continental Company (MCC) dated February 4, 2000 that omitted the use of screen tubes as a part of the installation procedure of the adhesive anchoring system (See Attachment 31 – Contractor transmittal from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 4, 2000). Listed below are excerpts from the contractor transmittal:

“.....Work May Proceed Subject To The Changes Indicated, Resubmittal Not Required.

Section 3 – Installation Procedure & Quality Control Program
Adhesive Anchoring System

5. *On floor applications, the drill hole shall be filled approximately one half full with the epoxy adhesive, starting from the rear of the hole.*

On wall and ceiling applications, after the hole is drilled and cleaned, insert the proper size anchor seal plug into the hole until the rim of the plug is fully seated against the base material. Insert the mixing nozzle through opening in the seal plug, and fill the drill hole approximately one half full, starting from the rear of the hole.....”

The work to install the ceiling system in the I-90 eastbound tunnel (from November 15, 1999 to March 13, 2000) began prior to the omission of screen tubes in the third contractor transmittal (submittal number 723.480-008 D) sent by Bechtel / Parsons Brinckerhoff (B/PB) to Modern Continental Company (MCC) dated February 4, 2000.

14. INSPECTION OF CEILING SYSTEM

No documentation was provided to NTSB investigators that inspections were performed to determine the physical and functional condition of the ceiling system from the time the I-90 eastbound connector tunnel was opened to traffic, January 18, 2003, to the day of the collapse. Approximately 3 years and 7 months (or 43 months) passed in which no routine inspections were performed to the ceiling system in the I-90 eastbound connector tunnel.

14.1 SCOPE OF SERVICES FOR A STRUCTURAL INSPECTION PROGRAM

A scope of services was contained in Work Program 13²³ and Work Program 16²⁴ for Bechtel / Parsons Brinckerhoff (B/PB) to develop a Structural Inspection Program.

“The CONSULTANT will develop a maintenance structural inspection program for the viaducts, tunnels, bridges, buildings and other facilities constructed by MHD as part of the CA/T Project and transferred to the AUTHORITY. The scope

²³Work Program 13, Amendment #22, Scope of Services, Section 7.4.2 Structural Inspection Program, page 26 of 27, January 18, 2002.

²⁴Work Program 16, Scope of Services, Section 3.3.8 Structural Inspection Program, page 24.

of the program will deal with the structural elements of the tunnels, tunnel finishes, bridges, buildings and facilities as well as the structural components of the electrical, mechanical and control systems and specialty products and materials used in the construction of the CA/T Project. The inspection program will include procedures and protocols for routine and emergency inspections. The CONSULTANT will develop maintenance and inspection manuals for Project segmental bridges and the Leonard Zakim Bunker Hill cable-stayed bridge. The CONSULTANT will develop a Geographic Information System (GIS) for inspection tracking and data recording.”

14.2 INSPECTION MANUAL FOR TUNNELS AND BOAT STRUCTURES PREPARED BY BECHTEL / PARSONS BRINCKERHOFF (B/PB)

Bechtel / Parsons Brinckerhoff (B/PB) prepared an inspection manual entitled Inspection Manual for Tunnels and Boat Structures dated November 2003. The purpose of the inspection manual was the following²⁵:

“The purpose of the tunnel inspection manual is to aid the Massachusetts Turnpike Authority (MTA) in the detailed and scheduled inspection of the tunnels, boat sections and the structural elements included within the Central Artery / Tunnel (CA/T) network. Tailored to this highway tunnel system, the scope encompasses all aspects of a structural inspection.”

One integral component of the inspection manual was the following²⁶:

“Section 5.0, Classification of Structure Type and Specific Elements, illustrates specific elements encompassed within the CA/T which is comprised of five different tunnel types. This mainly includes special structures, products and testing, concrete and steel components, protective coatings, tunnel finishes and the roadway wearing course.”

The manual included a set of condition ratings that applied to the structural elements of the inspection²⁷:

“In addition, a set of condition ratings, ranging from 0 to 9, can be applied to the structural elements of the inspection. These techniques assist in providing a straightforward inspection record that is effective for tunnel assessment and maintenance.”

²⁵Inspection Manual for Tunnels and Boat Structures, Volume 5, prepared by Bechtel / Parsons Brinckerhoff, November 2003, page 1.

²⁶Inspection Manual for Tunnels and Boat Structures, Volume 5, prepared by Bechtel / Parsons Brinckerhoff, November 2003, page 1.

²⁷Inspection Manual for Tunnels and Boat Structures, Volume 5, prepared by Bechtel / Parsons Brinckerhoff, November 2003, page 1.

The manual indicated that each hanger component used to support the modular concrete ceiling panels in the I-90 and I-93 exhaust duct areas should be inspected visually, by sounding or by nondestructive testing²⁸:

“The ceiling hanger shown in Figure 5.2.4-2 is a typical unit used to support the modular concrete ceiling panels in all other areas of I-90 and I-93 exhaust duct construction. Previously referenced in Section 5.1.4 (2e – Hanger Support System), each hanger component shall be inspected visually, by sounding or by nondestructive testing.”

Appendix A of the manual included a blank routine hanger inspection field report. The blank field report identified entry fields that included hanger no., location, connections for ceiling slab and floor slab, clevis, bolts, turnbuckle, threaded rod, moisture, and comments. Appendix A also included a deficiency legend that identified the following deficiencies for exhaust duct hangers (vertical and diagonal) and bolt connections: surface rust, loss of section %, loss of tension, out of plane (horizontal and vertical), broken, and buckled.

14.2.1 STATUS OF INSPECTION (NOVEMBER 2003 – JULY 2006)

The Massachusetts Turnpike Authority (MTA) provided a summary of the status of the Inspection Manual for Tunnels and Boat Structures from November 2003 through July 2006 (See Attachment 32 – Memorandum from the Massachusetts Turnpike Authority (MTA) to NTSB dated January 12, 2007). The memorandum indicated the following:

“A. November 2003 – July 2006

B/PB submitted the Inspection Manual for Tunnels and Boat Structures (the Manual) to the MTA in November 2003. Upon receipt of the Manual and as CA/T construction progressed, MTA continued to work with B/PB and FHWA to develop a tunnel inspection program based on the Manual. This effort included discussions regarding the Tunnel Inspection Database, a training program for inspection personnel and Bridge (Tunnel) Identification Number mapping. In late 2005, MTA took steps to initiate pilot or trial inspections based on the Manual, including engagement of an outside consultant. However, these trials were never carried out.”

14.2.2 STATUS OF INSPECTION (POST JULY 2006)

The Massachusetts Turnpike Authority (MTA) provided a summary of the status of tunnel inspections after the day of the collapse, post July 2006 (See Attachment 32 – Memorandum from the Massachusetts Turnpike Authority (MTA) to NTSB dated January 12, 2007). The memorandum indicated the following:

“B. Post July 2006

²⁸Inspection Manual for Tunnels and Boat Structures, Volume 5, prepared by Bechtel / Parsons Brinckerhoff, November 2003, page 111.

Since July 2006, a number of inspection efforts have been underway throughout the MHS, including a “Stem-to-Stern” Safety Review. Findings from Phase 1 of this review were recently submitted to MTA and it is expected that two additional phases of review are to follow. Applicable recommendations from this three-phase review will be incorporated into future maintenance and inspection efforts. In addition, MTA has reinvigorated efforts to develop and implement a comprehensive tunnel inspection program, including:

- MTA recently issued a Tunnel Inspection and Testing Program Policy Directive that calls for routine, in-depth inspections of all tunnel elements. This policy (attached) requires the inspection of all appurtenances suspended over roadways and their hanger systems to be inspected annually and all other structural tunnel components to be inspected every 3 years. Specific inspection protocol for the MHS system will be developed and fine-tuned throughout the inspection process based on the Manual and the FHWA/FTA Highway and Rail Transit Tunnel Inspection Manual.*
- In December 2006, MTA issued a Request for Proposals (attached) for a General Engineering Consultant to assist in the development of a comprehensive tunnel inspection program, including the completion of a tunnel inspection database, a training program for inspection personnel and comprehensive inspection procedures and protocol. MTA expects to begin work on the tunnel inspection program with the selected consultant(s) this spring.*
- MTA is currently reorganizing its engineering divisions to address the growing needs of the MHS more effectively. In particular, the new organization includes an MHS engineering division that will report to an Assistant Chief Engineer for the MHS. This new alignment allows for a more specialized staff to address the particular needs and complexities of the MHS. As it relates to the inspection program, this reorganization calls for a newly created tunnel and bridge inspection unit within MHS engineering with six new full-time inspection personnel to work with outside consultants to oversee and implement a rolling inspection program.*
- MTA is developing a five-year capital program that will incorporate a robust tunnel inspection program, as required by the MTA Tunnel Inspection and Testing Program Policy.”*

14.3 NATIONAL BRIDGE INSPECTION PROGRAM

The U.S. Department of Transportation Federal Highway Administration (FHWA) had the legislative authority under the Code of Federal Regulations (23 CFR Part 650) to develop a

national bridge inspection program. No such legislative authority exists in the CFR to develop a national tunnel inspection program. The CFR indicated the following:

“650.301 Purpose.

This subpart sets the national standards for the proper safety inspection and evaluation of all highway bridges in accordance with 23 U.S.C. 151.

650.307 Bridge inspection organization

(a) Each State transportation department must inspect, or cause to be inspected, all highway bridges located on public roads that are fully or partially located within the State’s boundaries, except for bridges that are owned by Federal agencies.”

The national bridge inspection program was formed as a direct result from a bridge collapse that occurred in Point Pleasant, West Virginia on December 15, 1967 that killed 46 people. The tragic collapse aroused national interest in the safety inspection and maintenance of bridges when a 2,235-foot section of the Silver Bridge collapsed into the Ohio River. This was one of the first accidents that NTSB investigated (Report Number HAR-71-01) with a recommendation to the Secretary of Transportation to develop inspection equipment capable of detecting critical flaws in standing bridge structures and devising analytical procedures to identify critical locations in bridge structures which require detailed inspection (Recommendation Number H-71-013).

The national bridge inspection program consists of national bridge inspection standards (NBIS) and a national bridge inventory (NBI). The national bridge inspection standards (NBIS) were first established in 1971 to set national requirements regarding bridge inspection frequency, inspector qualifications, report formats, and inspection and rating procedures. The national bridge inventory (NBI) is the aggregation of structure inventory and appraisal data collected by each state to fulfill the requirements of the program. The structure inventory data consists of fields that include identification of the bridge, structure type and material, age and service, geometric data, navigation data, and classification. The structure appraisal data consists of fields that include condition, load rating and posting, appraisal, proposed improvements, and inspections.

The national bridge inspection program requires bridges be inspected at regular intervals not to exceed 24 months.

“650.311 Inspection frequency

(a) Routine inspections. (1) Inspect each bridge at regular intervals not to exceed twenty-four months.”

Bridge inspectors are required to be trained regarding proper bridge inspection techniques and complete a Federal Highway Administration (FHWA) approved comprehensive bridge inspection training course.

The Federal Highway Administration (FHWA) developed a manual entitled “Highway and Rail Transit Tunnel Inspection Manual 2005 Edition”²⁹ that established the frequency for inspections of tunnel structures:

“The tunnel owner should establish the frequency for up-close inspections of the tunnel structure based on the age and condition of the tunnel. For new tunnels, this time period could be as great as five years. For older tunnels, a much more frequent inspection time period may be required, possibly every two years. This up-close inspection is in addition to daily, weekly, or monthly walk-through general inspections.”

The “Highway and Rail Transit Tunnel Inspection Manual 2005 Edition” is a guide for tunnel inspections. The recommendations in the manual do not carry any legislative authority.

15. AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

The American Association of State Highway and Transportation Officials (AASHTO) is a nonprofit, nonpartisan association founded in 1914 as the American Association of State Highway Officials. It represents highway and transportation departments in the 50 States, the District of Columbia, and Puerto Rico. The association’s mission is to advocate transportation-related policies and to provide technical services to support States in their efforts to efficiently and safely move people and goods.

AASHTO is a preeminent source of technical information on highways and other transportation facilities. AASHTO conducts extensive research and data analysis to provide written guidelines and standards on particular transportation topics.

AASHTO members consist of representatives from various state transportation departments that serve on subcommittees and technical committees including representatives from the Federal Highway Administration (FHWA). The committees that deal with tunnels are the Subcommittee on Bridges and Structures and the Tunnels (T-20) technical committee. The Tunnels (T-20) technical committee would be charged with developing a tunnel design manual should legislative authority be approved for a national tunnel inspection program. The tunnel design manual would entail design, construction, and inspection standards for tunnels and tunnel finishes. The next meeting of the Subcommittee on Bridges and Structures and the Tunnels (T-20) technical committee is scheduled for July 8-12, 2007 in Wilmington, Delaware.

FHWA developed a manual entitled “Road Tunnel Design Guidelines”³⁰ in July of 2004 that contained guidelines for the design of different categories of tunnels (soft ground tunneling, rock tunnels, mixed-face and difficult ground tunnels, shafts, shotcrete, immersed tunnels, and cut-and-cover tunnels). The manual is a guide for the design of tunnels and does not contain any

²⁹Highway and Rail Transit Tunnel Inspection Manual, 2005 Edition, Federal Highway Administration and Federal Transit Administration, Chapter 4: Inspection Procedures – General Discussion, page 4-1.

³⁰FHWA Road Tunnel Design Guidelines, Federal Highway Administration, Report No. FHWA-IF-05-023, July 2004.

guidance on the design of tunnel finishes. The recommendations in the manual do not carry any legislative authority.

16. PROPOSAL BY GANNETT FLEMING, INC. FOR ALTERNATE ANCHOR SYSTEM

Gannett Fleming, Inc. sent a letter to Bechtel / Parsons Brinckerhoff (B/PB), the owner's authorized representative, dated August 25, 1997 (See Attachment 33 – Letter from Gannett Fleming, Inc. to Bechtel / Parsons Brinckerhoff (Authorized Representative) dated August 25, 1997). The letter indicated that Gannett Fleming, Inc. had researched various types of post-installed anchoring systems for hanging the ceiling in the C04A2 portion of the C09B2 tunnel finishes contract. The various types of post-installed anchorage systems included the resin bonded anchor, the undercut anchor, and the expansion anchor. The letter mentioned the undercut anchor appeared to have the desired characteristics for supporting the ceiling system in the subject tunnel. The letter indicated the following:

“UNDERCUT ANCHORS

Undercut anchors consist of a threaded rod with nut assembly, expansion sleeve and expansion plug. When the nut is tightened, the expansion plug expands the steel expansion sleeve into the undercut at the top of the drilled hole. Tensile forces are transferred into the concrete by keying and bearing against the undercut in the concrete.

A research report which we reviewed stated that an undercut anchor in cracked concrete will retain 65% to 70% of the anchors ultimate capacity in uncracked concrete. Undercut anchors perform the same as cast-in headed anchors according to this paper. These anchors will perform similar to steel when exposed to high temperatures.

Undercut anchors are recommended by manufacturers for application in the tensile zone of concrete as well as cracked concrete. They are recommended for dynamic loading as well. These anchors are available in a variety of steel materials and with zinc plating or hot-dipped galvanized coatings depending on the manufacturer. These anchors appear to have the desired characteristics for supporting the ceiling system in the subject tunnel.”

17. MEMORANDUM CITING BECHTEL / PARSONS BRINCKERHOFF (B/PB) DIRECTION TO USE A FACTOR OF SAFETY OF TWO (2)

Gannett Fleming, Inc. prepared a memorandum dated October 12, 1998 (See Attachment 34 – Memorandum from Gannett Fleming, Inc. dated October 12, 1998) that summarized a meeting in which Bechtel / Parsons Brinckerhoff (B/PB) had given direction to Gannett Fleming, Inc. to use a factor of safety³¹ of two (2) toward the manufacturer's ultimate capacities³² to arrive at allowable capacities³³ for adhesive design. The memorandum indicated the following:

³¹Factor of safety is the ratio of predicted ultimate load to the calculated maximum service load.

“This meeting focused on the use of adhesive anchors which will be used to attach the hanging ceiling supports to the concrete tunnel roof.

P. Das directed us to incorporate the following design criteria into our present design.

1. *For the one-hanger out scenario:*
 - a. *use a factor of safety of two (2) applied to manufacturer’s ultimate capacities to arrive at allowable capacity for adhesive design.”*

18. METROPOLITAN HIGHWAY SYSTEM “STEM TO STERN” SAFETY REVIEW – PHASE 1

In response to emergency legislation, Governor Mitt Romney issued Executive Order 474 establishing the Metropolitan Highway System (MHS) Safety Review Team. The executive order directed the Safety Review Team to undertake a “stem to stern” safety review of the MHS. It further directed the team to establish an appropriate work plan to ensure that the most safety-critical elements of the MHS tunnels and facilities were given priority, and to report to the Governor on these priority elements within ninety days. The Safety Review Team prepared an eleven-volume series of reports entitled Stem to Stern Safety Review – Phase 1. The report contained a table that indicated the adhesive anchor systems in the MHS tunnel ceilings³⁴:

Tunnel (Year Ceiling Installed)	Ceiling Panels		Ceiling Attachment		Adhesive Anchors			
	Type	Wt.	No. of Hangers	Attachment Type	No. per Hanger	Dia	Design Embed	Adhesive
<i>I-90 Connector (1999- 2000)</i>	<i>4-inch normal weight concrete</i>	<i>50 psf</i>	<i>10,800 (Note 1)</i>	<i>Bolts in embedded steel channels, typical</i>				
			<i>1,300 (Note 1)</i>	<i>Adhesive anchors at east tunnel portals and isolated locations</i>	<i>2 or 4</i>	<i>5/8”</i>	<i>5 1/2”</i>	<i>Epoxy resin: Power- Fast</i>
<i>I-90 Ted Williams (1994)</i>	<i>2-inch composite lightweight</i>	<i>20 psf</i>	<i>12,200 (Note 2)</i>	<i>Adhesive anchors</i>	<i>2 typ</i>	<i>3/4” typ</i>	<i>5 1/2”</i>	<i>Epoxy resin: Ultra</i>

³²Ultimate load capacity is the maximum load that may be placed on a structure, causing failure of that component.

³³Allowable load capacity is the stress that can be sustained with acceptable safety by a structural component under the particular condition of service and loading.

³⁴Metropolitan Highway System Stem to Stern Safety Review – Phase I, Volume 3 – Tunnels – Finishes, November 15, 2006, Appendix B.

(1994-1995)	lightweight concrete core		4,000 (Note 3)					Ultra-Bond (Note 5)
I-93 Central Artery (2001-2002)	4-inch normal weight concrete	50 psf	12,800	Hanger plates are typically attached to steel roof girders				
			55 (Note 4)	Adhesive anchors at tunnel section under MBTA Red Line	2	1"	22"	Vinyl ester resin: Hilti HVU
Sumner Tunnel (1996-1997)	2-inch composite lightweight concrete core	15 psf	2,739 (Note 6)	Adhesive anchors (frequently installed in spalled and delaminated concrete)	1 typ	3/4"	6 5/8"	Polyester resin: KeligROUT 101
Callahan Tunnel (1991-1992)	2 1/4"-inch composite mineral board	15 psf	2,338 (Note 6)	Adhesive anchors (and steel welding)	1 typ	3/4"	6 5/8"	Epoxy resin: Foil-Fast Epoxy and Polyester Resin: Lokset Anchor-bond

Note 1 – I-90 Connector totals include Ramp DN ceilings installed by Contract C17AA.

Note 2 – Total includes 9,500 hangers, 2,700 diagonal braces.

Note 3 – Estimated number of anchors securing direct-attached ceiling framing to roof.

Note 4 – Number of upper hanger plate connections to concrete roof structure.

Note 5 – Product identified in approved contractor submittal; other products may have been used.

Note 6 – Total including diagonal braces.

D. ATTACHMENTS

- Attachment 1 – Memorandum to the National Transportation Safety Board from Gannett Fleming, Inc. dated August 4, 2006 (Updated November 2, 2006)
- Attachment 2 – Contractor transmittal (submittal number 723.480-014 C) from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000 regarding the contractor's anchor adequacy submittal certified by Professional Engineer (PE) seal
- Attachment 3 – Design Policy Memorandum No. 107 Tunnel Finish Details dated March 13, 1997
- Attachment 4 – Federal Highway Administration (FHWA) Turner-Fairbank Highway Research Center Finite Element Design Analysis Report
- Attachment 5 – Manufacturer literature from Power's Fasteners, Inc. on maximum torque values and torque ranges for the adhesive anchoring system
- Attachment 6 – Facsimile Transmittal from Massachusetts Highway Department to Gannett Fleming, Inc. dated September 23, 1998 giving direction on proof test load for anchors
- Attachment 7 – Contractor transmittal (submittal number 723.480-009 B) from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated October 15, 1999 regarding the ceiling module erection procedure
- Attachment 8 – Manufacturer literature from Power's Fasteners, Inc. on installation requirements for the adhesive anchoring system
- Attachment 9 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated October 7, 1999 indicating a small percentage of the adhesive anchors in the HOV ceiling mockup appear to show signs of tensile movement
- Attachment 10 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated October 12, 1999
- Attachment 11 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated November 8, 1999
- Attachment 12 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated November 11, 1999
- Attachment 13 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated December 2, 1999

- Attachment 14 – Contractor transmittal (submittal number 723.480-016 A) from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated December 27, 1999 indicating the proposed fix for the failed anchor bolts, as submitted, is rejected
- Attachment 15 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated December 29, 1999
- Attachment 16 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000
- Attachment 17 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 7, 2000
- Attachment 18 – Response Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated January 19, 2000
- Attachment 19 – Response Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 7, 2000
- Attachment 20 – Adhesive Anchor Repair Procedure submitted by Modern Continental Company to Bechtel / Parsons Brinckerhoff dated February 16, 2000
- Attachment 21 – Internal Memorandum from Staff of Bechtel / Parsons Brinckerhoff dated August 31, 2000 Recommending Disposition of Deficiency Report No. 1 be Closed
- Attachment 22 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated September 18, 2000 requesting a contract modification be issued for reimbursement of the additional work associated with retesting the adhesive anchors to 6,350 pounds
- Attachment 23 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated January 26, 2001 indicating corrective action had been verified and Deficiency Report No. 1 was closed
- Attachment 24 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated March 15, 2001 enclosing a fully executed original of the contract modification
- Attachment 25 – The Commonwealth of Massachusetts Office of the Inspector General, A Review of the Central Artery / Tunnel Project's Use of Anchor Bolts on the C05B1 Tunnel Finishes Contract

- Attachment 26 – Letter from Modern Continental Company to Bechtel / Parsons Brinckerhoff dated January 13, 1999 requesting consideration of a Celline Metal Panel Structural Ceiling System
- Attachment 27 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 24, 1999 indicating Value Engineering Cost Proposal was not acceptable
- Attachment 28 – Letter from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated March 25, 1999 indicating Value Engineering Cost Proposal should not be pursued further and is considered not viable to implementation for the C09B2 project
- Attachment 29 – Letter from Environmental Interiors, Inc. to Modern Continental Company (MCC) dated March 9, 1999 withdrawing a proposal for a Celline Metal Panel Ceiling System
- Attachment 30 – Contractor transmittal (submittal number 723.480-008 A) from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated May 27, 1999 regarding the installation procedure of the adhesive anchoring system and the use of screen tubes
- Attachment 31 – Contractor transmittal (submittal number 723.480-008 D) from Bechtel / Parsons Brinckerhoff (Authorized Representative) to Modern Continental Company dated February 4, 2000 regarding the installation procedure of the adhesive anchoring system and omitting the use of screen tubes
- Attachment 32 – Memorandum from the Massachusetts Turnpike Authority (MTA) to NTSB dated January 12, 2007 regarding Inspection Manual for Tunnels and Boat Structures, Volume 5
- Attachment 33 – Letter from Gannett Fleming, Inc. to Bechtel / Parsons Brinckerhoff (Authorized Representative) dated August 25, 1997 regarding proposal for alternate anchor system
- Attachment 34 – Memorandum from Gannett Fleming, Inc. dated October 12, 1998 regarding the use of factor of safety of two (2) applied to manufacturer's ultimate capacities

Dan Walsh /s/

Dan Walsh
Highway Accident Investigator